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OECD

ECONOMIC SURVEYS

ORGANISATION FOR ECONOMIC CO-OPERATION AND DEVELOPMENT

GERMANY

ORGANISATION DE COOPÉRATION ET DE DÉVELOPPEMENT ÉCONOMIQUES

APRIL 1970

BASIC STATISTIC OF GERMANY (F.R.)

THE LAND

Area, 1968 (thousand km²)	249
Agricultural area, 1968 (thousand km²)	139
Forests, 1968 (thousand km²)	72

Major cities, June 1968	Million inhabitants
Berlin (West)	2.2
Hamburg	1.8
Munich	1.3
Cologne	0.9
Essen	0.7
Dusseldorf	0.7
Frankfurt	0.7
Dortmund	0.6

THE PEOPLE

Population 1968 (thousand)	60 184
No. of inhabitants per km²	242
Net natural increase in population (annual average 1966-1968)	311 000
per 1 000 inhabitants	5.2
Net immigration (annual average 1966-1968)	78 000
Labour force, total 1969	26 822 000
Labour force in industry, April 1969	12 741 000

PRODUCTION

GNP 1969 (billions of DM)	601.4
GNP per head 1968 (US $)	2 240
Gross fixed investment (average 1966-1968):	
per cent of GNP	24
per head (US $)	510
Origin of GDP, 1969 (per cent):	
Agriculture, forestry, fishery	4
Industry (incl. construction)	53
Services	43
Home food production as a percentage of total food availability (agricultural year 1966-1967)	63

THE GOVERNMENT

Public consumption 1969 (per cent of GNP)	15.5
General government current revenue 1968 (per cent of GNP)	37
Public debt, 1968 (ratio to general government current revenue)	59

Composition of Federal Parliament

Christian Democrats	242	seats
Social Democrats	224	—
Free Democrats	30	—

Last election: 28th September 1969.
Next election: 1973.

LIVING STANDARDS

Calories per head, per day 1967-1968	2 900
Average gross hourly earnings of male industrial workers, 1969 (DM)	5.63
No. of passenger cars in use, mid-1968 (per 1 000 inhabitants)	194
No. of telephones end 1967 (per 1 000 inhabitants)	172
Energy consumption per head 1968 (kg oil equivalent)	3 370
No. of television sets end 1968 (per 1 000 inhabitants)	248
Public current expenditure on education per head, 1968 (US $)	50

FOREIGN TRADE

Exports:	
Exports of goods and services as per cent of GNP (average 1967-1969)	22
Main exports 1969 (per cent of total merchandise exports):	
Products of agriculture, forestry and fishing	1
Basic materials and semi-finished goods	29
Manufactured foods and tobacco	3
Other consumer manufactures	11
Investment goods	54
Other exports	2
Total	100

Imports:	
Imports of goods and services as per cent of GNP (average 1967-1969)	20
Main imports, 1969 (per cent of total merchandise imports):	
Products of agriculture, forestry and fishing	15
Basic materials and semi-finished goods	41
Manufactured foods and tobacco	8
Other consumer manufactures	13
Investment goods	19
Other imports	4
Total	100

THE CURRENCY

Monetary unit: Deutsche Mark.

Currency units per US dollar 3.66

NOTE. Figures include the Saar and West Berlin.

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GERMANY

ORGANISATION FOR ECONOMIC CO-OPERATION AND DEVELOPMENT

The Organisation for Economic Co-operation and Development was set up under a Convention signed in Paris on 14th December 1960 by the Member countries of the Organisation for European Economic Co-operation and by Canada and the United States. This Convention provides that the OECD shall promote policies designed:

- *to achieve the highest sustainable economic growth and employment and a rising standard of living in Member countries, while maintaining financial stability, and thus to contribute to the development of the world economy;*
- *to contribute to sound economic expansion in Member as well as non-member countries in the process of economic development;*
- *to contribute to the expansion of world trade on a multilateral, non-discriminatory basis in accordance with international obligations.*

The legal personality possessed by the Organisation for European Economic Co-operation continues in the OECD, which came into being on 30th September 1961.

The members of OECD are: Austria, Belgium, Canada, Denmark, Finland, France, the Federal Republic of Germany, Greece, Iceland, Ireland, Italy, Japan, Luxembourg, the Netherlands, Norway, Portugal, Spain, Sweden, Switzerland, Turkey, the United Kingdom and the United States.

The Socialist Federal Republic of Yugoslavia is associated in certain work of the OECD, particularly that of the Economic and Development Review Committee.

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April 1970

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INTRODUCTION

The past year has seen demand pressures becoming increasingly serious, and costs and prices rising fast. The balance of the economy suffered from the delays which intervened before essential policy changes were undertaken—both the decision to revalue the currency and to moderate domestic demand. The prospects are now for some easing of demand pressures and for progress towards a more satisfactory price performance. But with demand management heavily dependent on monetary restraint, uncertainties are attached to the prospects for balanced development, both on the upward and the downward side. These prospects—together with their implications for policy—are discussed in Part III of the present Survey. Part III also considers the prospect for the balance of payments: in particular the question of how far a falling surplus on current account might be accompanied by a strengthening of the capital account. The Survey seeks to look ahead to the middle of 1971.

I THE CURRENT SITUATION

Strong Demand Growth

In 1969, real GNP rose by some 8 per cent, even more than in 1968 (7¼ per cent). With capacity utilisation rising rapidly in the course of 1968, the economy entered 1969 with considerable momentum. In the first half of the year weather-induced stagnation in construction and the unwinding of anticipatory effects on exports related to the introduction of the border taxes in late 1968 served to reduce the growth of activity. But with business investment and private consumption rising fast, real GNP increased at an annual rate of some 6½ per cent. By the middle of the year, demand pressures were much stronger than during the peak of the 1964-65 boom, with capacity utilisation in industry at a record level and unemployment very low. Nevertheless, in the second half, real GNP rose at an annual rate of about 8 per cent, helped by a sharp rise in the foreign labour force; between March and September the number of foreign workers rose by 270 000, or more than 1 per cent of the dependent labour force.

The continued strong expansion took most observers by surprise. Forecasts published at the beginning of the year ranged from 3¼ to 5 per cent, and most forecasters felt that there was a risk of undue slowdown in the second half[1]. The reasons given for the envisaged slower growth in 1969 were generally as follows:

(*a*) German export markets were thought likely to grow significantly less than in 1968, mainly because of the expected slowdown of the expansion of home demand and imports in certain important countries, notably the UK and the US. It was also assumed that stronger demand and cost pressures would adversely affect export performance. Demand in the UK and the USA was, however, rather stronger than expected, and the imports of France, Germany's most important customer, expanded very strongly indeed;

(*b*) The IFO-test of September 1968 pointed to a slower rise in business fixed investment in the second half of the year. Such a development seemed plausible given the likelihood of weaker export growth and the fact that business fixed investment in 1969 was likely to reach a level high in relation to its long-term trend;

(*c*) No further stimulus was to be expected from inventory accumulation in view of the strong build-up which had taken place in 1968.

Table 1 compares the OECD forecast with the actual outcome. It can be seen that, in terms of ex post arithmetic, most of the difference was due to greater buoyancy of domestic demand including stockbuilding. The stronger-than-expected foreign balance " explains " only one half of one per cent of GNP. But this sort of calculation greatly underestimates the importance of the foreign balance. The fact that the volume of exports of goods and services rose by 12 per cent rather than the 7 per cent envisaged represented additional demand equivalent to almost 1¼ per cent of GNP[2]. Even though much of this, as far as GNP impact is concerned, was offset by a stronger-than-expected rise in imports, the sharp rise in exports clearly had an important influence on the business climate, and thus on the spend-

1 The German Institutes of Economic Research were at that time forecasting the following annual increase in real GNP: IFO 3.2 per cent, Kiel 3.3 per cent, Berlin, Essen and Hamburg 4 per cent. The Annual Economic Report of the Government forecast a rise of 4.5 per cent, and the Council of Economic Advisers (November 1968) 4.7 per cent (without allowing for effects of revaluation or border taxes). Even in the late spring of 1969 (25.4.1969) 5 of the 6 German economic research institutes predicted a (spontaneous) weakening of the boom in the second half of 1969. At that time the Institutes were forecasting a GNP rise of 5.5 per cent. According to 5 of the 6 Institutes, the restrictive demand management measures taken or envisaged at that time by the Federal Government were likely to aggravate the weaker trend in domestic demand which would result from reduced investment activity in the second half of the year. By mid-year the IFO-institute was still maintaining its 5.5 per cent forecast for 1969 (OECD 6.5 per cent).

2 This measurement does not take into account secondary effects.

Table 1 **Demand and Output**
Comparison of Forecast[1] and Outcome

	1968 DM Billions Current prices	1969 OECD Forecast[1] percentage changes (volume)	1969 Outcome percentage changes (volume)
Private consumption	301.1	5	7¾
Public consumption	83.6	4	4
Fixed investment	124.8	9	12½
(Construction)	71.9	4	3½
(Equipment)	52.9	16	24¾
Final domestic demand	509.5	5¾	8¼
Plus change in stock building[2]	11.0	(0)	(½)
Total domestic demand	520.5		
(Exports)	124.4	7¼	(12)
(Imports)	106.4	12	(15½)
Plus change in foreign balance[2]	18.0	(−¾)	(−¼)
GNP	538.5	5	8

1 Economic Outlook No. 4, December 1968.
2 The changes in stock building and in foreign balance are expressed as a percentage of GNP in the previous period. These two figures plus the rate of change of final domestic demand broadly equal the rate of change of GNP.

ing decisions of both investors and consumers. Investment plans were revised upwards in the course of the year, so that real fixed asset formation rose by 12½ per cent rather than the 9 per cent forecast. This, in turn, together with the stronger foreign balance, generated higher employment, personal incomes and consumer expenditure. The failure to forecast exports correctly—which was closely related to the fact that the United States' merchandise imports rose by 8½ per cent rather than the 2½ per cent forecast—was therefore also a key element in the underestimation of the strength of domestic demand.

Production, employment and productivity

The acceleration of the rise in GNP was attributable entirely to stronger trends in industry and in trade and communications. The growth rate of output in agriculture and forestry dropped somewhat, and the expansion in the service sectors remained at the relatively modest rate of the previous year.

In 1969 it was official policy to achieve and maintain the rate of capacity utilization experienced in 1965. But with the strong advance in output,

Table 2 **Percentage Changes in Production by Sector 1966-1969**
Volume changes over previous year
1962 prices

	Weight in GDP 1969	1966	1967	1968	1969
Agriculture and forestry	3.6	2.5	10.2	4.3	2.4
Industry[1]	53.9	2.5	–2.4	9.5	10.4
Trade and communications	18.4	2.2	0.1	5.1	7.9
Services	13.8	4.7	3.0	2.6	3.0
Public authorities and private households	10.3	3.9	1.0	3.5	1.5
GDP	100.0	2.8	–0.3	7.0	7.9

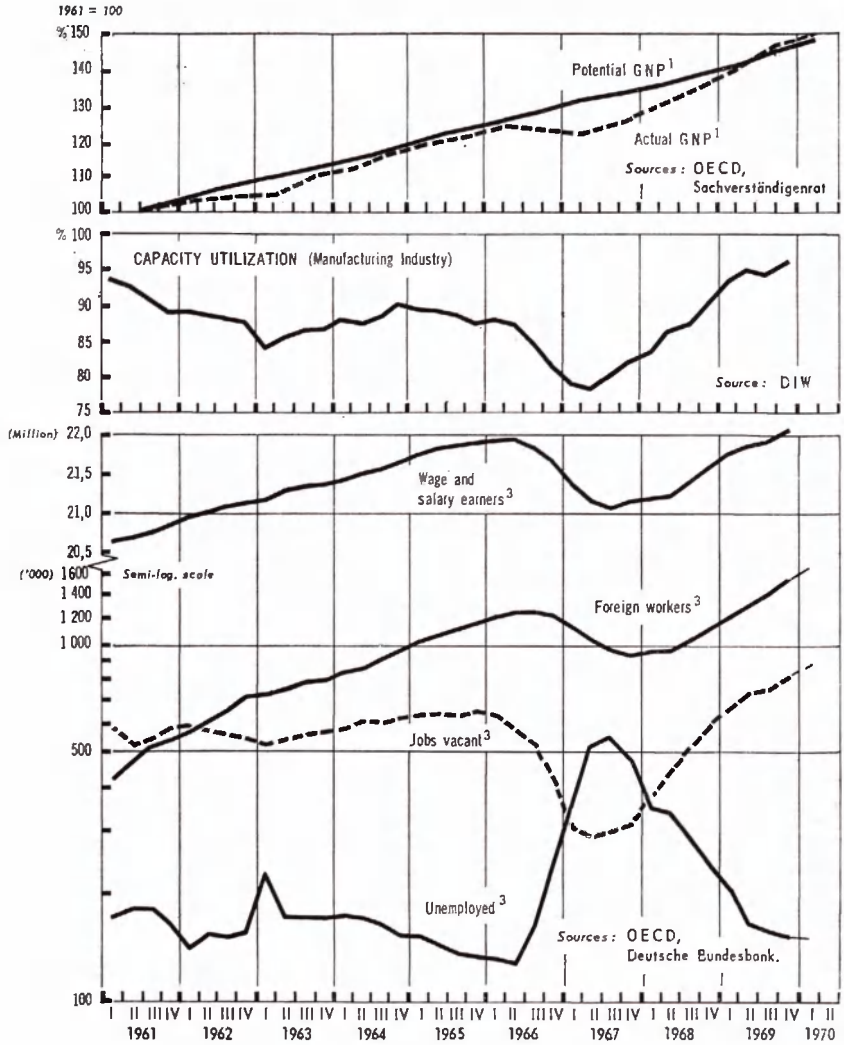
1 Including construction and public utilities.
Source: Statistisches Bundesamt.

the economy apparently exceeded the very high degree of resource utilization reached during the 1960 boom when demand pressures had been clearly excessive. During 1969 the order backlog in industry continued to rise rapidly, exceeding in December the high level of the corresponding month in 1960 (see Diagram 1).

The sharp rise in the rate of capacity utilisation was made possible by the unexpected elasticity of the labour market. By the end of 1969, the number of foreign workers had risen by some 400 000 to a post-war record level of about 1 600 000. The number of Germans employed also rose faster than earlier. Large numbers of persons marginal to the labour force were recruited, and by the end of the year virtually the whole of that part of the labour force which ceased work during the recession without registering as unemployed seems to have been re-integrated. Stronger pressures on the labour market also led to a sharp increase in overtime work. By the end of 1969 the seasonally adjusted unemployment rate had declined to the very low rate of 0.7 per cent. There were six jobs vacant for each unemployed person; in key sectors (e.g. metalworking) the proportion was even higher. Female part-time work continued, however, to play a relatively unimportant role. It would appear that further efforts to use the considerable potential of female part-time workers could help in future to reduce strain on the labour market.

The large increase in the foreign labour force was accompanied by a marked change in nationality structure (see Diagram 2). Most of the new recruitment took place in Turkey and Yugoslavia, whereas the number of Italians and Spaniards did not regain the pre-recession level. These developments were probably related to the decline in unemployment and the

Diagram 1 Production, Capacity and Employment



1 Constant prices.
2 Deviations of quarterly seasonally adjusted values from Secretariat estimate of trend.
3 Seasonally adjusted.

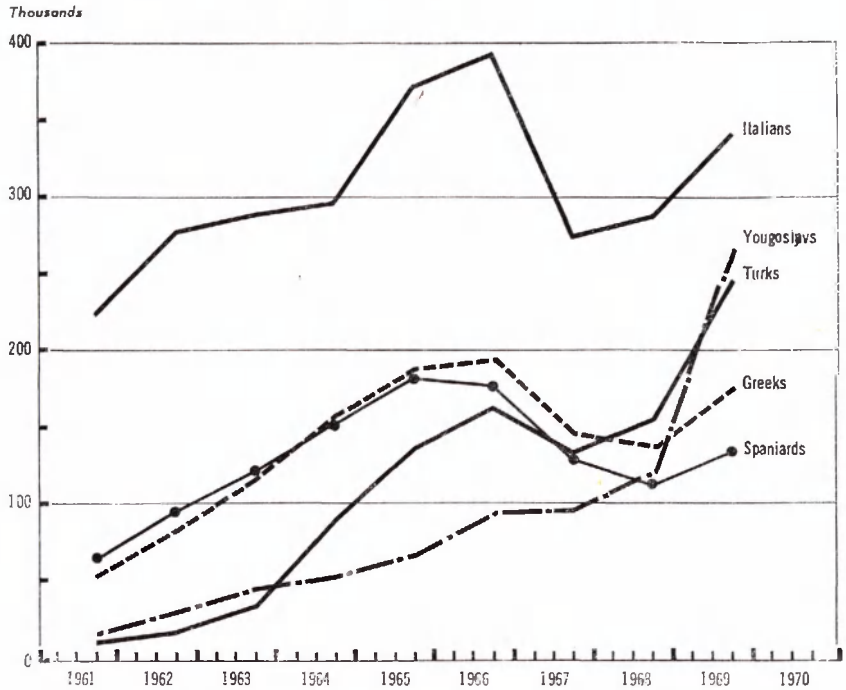
Table 3 **Labour Force**
Thousands, seasonally adjusted

	1966	1967		1968		1969	
	I	I	II	I	II	I	II
1 Labour force (2 + 3)							
Germans	26 080	25 820	25 740	25 590	25 670	25 655	25 595
Foreigners	1 240	1 075	965	970	1 065	1 260	1 470
Total	27 320	26 895	26 705	26 560	26 735	26 915	27 065
2 Employment							
Independent	5 230	5 130	5 095	5 025	5 000	4 935	4 880
Wage and salary earners	21 960	21 265	21 100	21 185	21 475	21 795	22 030
Total	27 190	26 395	26 195	26 210	26 475	26 730	26 910
3 Registered unemployed	130	500	510	350	260	185	155
4 Labour reserve[1]	—	255	345	300	200	100	60
5 Total employable reserve (3 + 4)	130	755	855	650	460	285	215
6 5 as per cent of 1	0.5	2.8	3.2	2.4	1.7	1.1	0.8

1 The "labour reserve" consisted of wage earners who withdrew from the labour force during 1967 without registering as unemployed. The estimates in Table 3 are based on a publication of the Institut für Arbeitsmarkt und Berufsforschung.

Sources: Institut für Arbeitsmarkt und Berufsforschung Mitteilungen Nr. 9 and Secretariat estimates.

Diagram 2 Trends in Nationality Structure of Foreign Labour Force[1]



1 End-September Figures.
Source: Bundesanstalt für Arbeit, Nürnberg.

sharp increase in wages in Italy and Spain since 1965-66. It remains to be seen to what extent the DM revaluation will increase the attraction of the German labour market, since foreign workers repatriate a large part of their earnings.

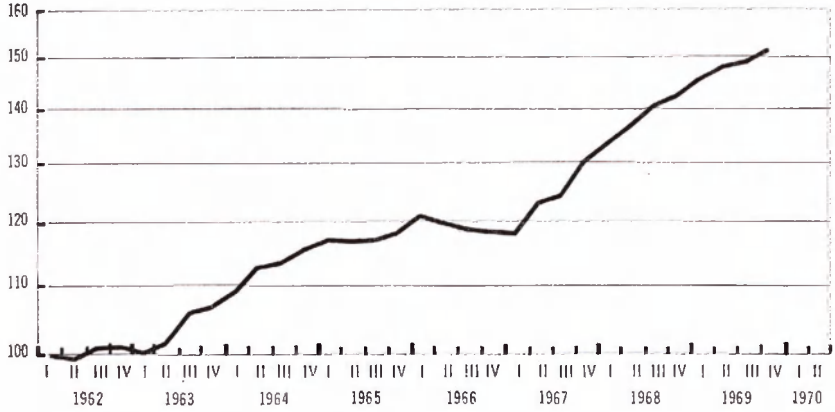
The industrial structure of employment has also changed somewhat. By the second half of 1969 total industrial employment was roughly back to the pre-recession level but the relative share of manufacturing, notably the investment goods industry, had increased at the expense of construction (see Table 4). During the whole of 1969 building activity was severely hampered by labour bottlenecks which aggravated the loss of production due to of bad weather conditions, and the rise in real output was modest. The reluctance of the labour force to return to the construction sector may have been due to the 1966-67 experience when construction jobs turned out to be especially unstable, and the relatively faster rise of hourly earnings in other sectors in 1968 and early 1969.

Table 4 **Structure of industrial employment**

	1966	1969
Manufacturing	82.9	84.2
Basic materials	18.7	18.4
Investment goods	37.7	39.4
Consumer goods	21.1	21.0
Food, drink, tobacco	5.4	5.4
Construction	17.1	15.8
Total industry	100.0	100.0
Number of employed persons (thousands)	9 629	9 496

Source: Wirtschaft und Statistik.

Diagram 3 **Productivity per Employed Person in Industry**
(1962 = 100, seasonally adjusted)



Source: Deutsche Bundesbank.

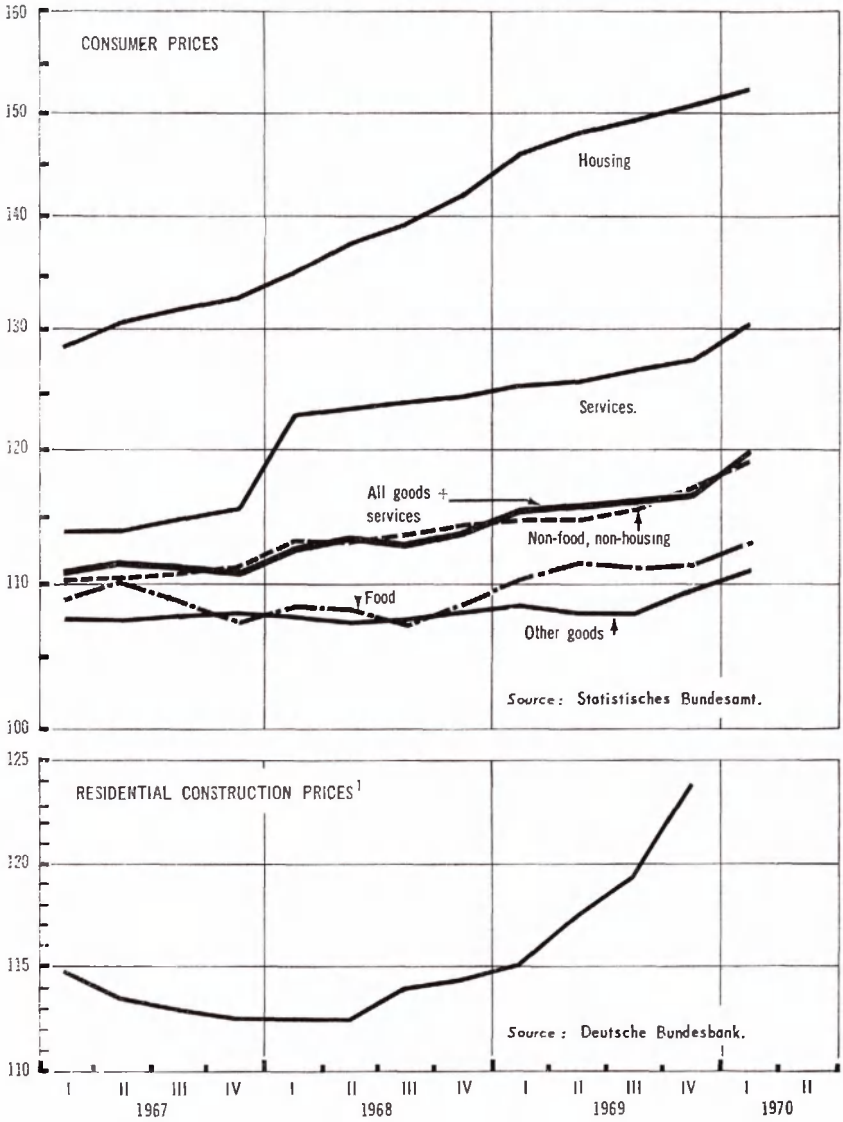
In contrast to previous cyclical experience the strong increase in labour productivity in industry continued unabated until the end of 1969 (see Diagram 3). The following factors may have contributed to this development :

(*a*) Strong demand has been associated not only, as in previous booms, with a more intensive utilization of existing physical capacities but—reflecting the tight labour market situation—also with a better utilization of the fixed part of the labour force;

(*b*) Entrepreneurs seem to have concentrated on investment with a high rationalisation effect;

(*c*) Finally, accelerated movement of workers from the less productive sectors of the economy appears to have produced a once for all productivity gain.

Diagram 4 **Price Developments 1967-1970**
(1963 = 100)



1 Estimated effect of transition to TVA in 1968 eliminated.

Prices, Wages and Income Distribution

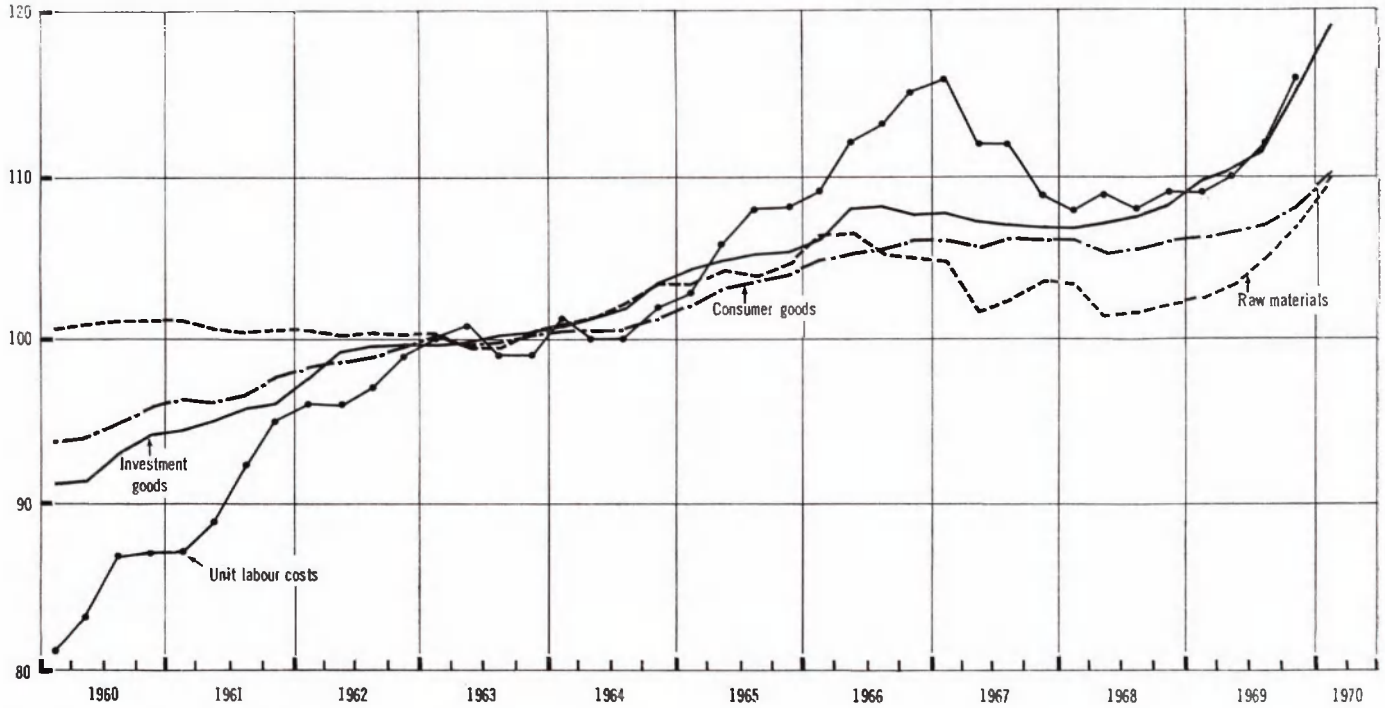
In spite of growing demand pressures, price increases remained moderate until mid-1969. Year over year the GNP deflator increased by 3.4 per cent in 1969 (1.5 per cent in 1968), but during the second half of the year the increase accelerated to an annual rate of some 4½ per cent. The cost of living index, stable until mid-1968, accelerated somewhat during the second half of 1968 and the first months of 1969 but this was largely attributable to the behaviour of food and rent prices and thus not closely related to the cycle. In the second half, the general rise in prices remained relatively modest, but prices affected by conjunctural forces rose at an accelerating pace and during the last quarter the cost of living index, excluding food and rent rose, at an annual rate of some 6 per cent. The rise in industrial producer prices accelerated sharply towards the end of the year to a rate unprecedented since the Korean boom (see Diagram 5), reflecting notably the very sharp increase in prices of investment goods, which during the second half of the year rose at an annual rate of 7½ per cent. Much of the very rapid increase in industrial producer prices has not yet been reflected in higher consumer prices. Export prices also have risen sharply. During the first few months of the year the rise resulted largely from the 4 per cent border tax measure, introduced in November 1968, which by the spring was fully reflected in the export price index. The upward trend continued in the second half; even after the revaluation DM-export prices continued to rise. By the beginning of 1970 dollar export prices were some 16 per cent up on the previous year's level. This undoubtedly exaggerates the loss of competitiveness since the export price index has a relatively remote base year (1962) and thus does not reflect the important changes in the structure of German exports in recent years[1].

By contrast, DM-import prices have changed little from the end- 1968 level. During the first half of the year the rise in prices charged by foreign exporters was roughly compensated by the 4 per cent import subsidy. A further price reduction for the German importer resulted from the fact that, for several months prior to the change in the DM-parity at the end of October, foreign currencies were available at a considerable forward discount. In January, after the revaluation (taking into account abolition of the 4 per cent import subsidy), non-agricultural imported goods were some 4-5 per cent more expensive than a year earlier, implying more than a 10 per cent rise in dollar prices. Nevertheless imports had gained a certain competitive edge on domestic products.

Wage increases also remained relatively moderate up to mid-1969. During the first half of the year wage settlements generally covering a period

1 For further discussion see Annex II.

Diagram 5 Industrial Producer Prices and Unit Labour Costs
1963 = 100



Sources: Deutsches Institut für Wirtschaftsforschung. OECD.

of one year provided increases in hourly rates averaging some 6 per cent. With wage drift continuing, gross wages and salaries per employee grew at a seasonally adjusted annual rate of about 8.5 per cent but, until late summer, the continued strong increase in productivity kept unit labour costs stable. The relatively peaceful wage climate changed dramatically in September when a wave of wildcat strikes led to advanced contract re-negotiations for iron and steel workers and miners, providing pay increases between 10 and 12 per cent. During the fourth quarter this led to a general revision of existing contracts over a wide range of activities, so that total gross wages and salaries per employee were 12½ per cent higher than a year earlier; seasonally adjusted this implied that, by the fourth quarter, wages and salaries were 5 per cent above the average for the year.

Table 5 **Wages and Profits**
Per cent changes at annual rate

	1967		1968		1969	
	I	II	I	II	I	II
Income from property and entrepreneurship	–4.0	9.2	12.0	18.0	8.0	10.5
Wages and salaries	–2.0	2.0	8.0	10.5	11.5	15.0
Wages per employee	2.5	5.0	6.0	8.0	8.5	12.5

1 Adjusted for turnover tax rebate in 1968.
Source: Statistisches Bundesamt, Secretariat seasonal adjustment.

In 1968, as an aftermath to the recession which, for the first time since the early fifties, had produced a considerable drop in employment, the rise in wages remained moderate in relation to the expansion of economic activity, and profits rose sharply. During the first half of 1969 the rise in wages caught up with the profit rise, and in the second half it clearly exceeded the increase in profits. In the second half of 1969 the share of gross wages and salaries in national income was roughly back to the 1965 level.

Balance of Payments

The current account surplus declined to $ 1.8 billion in 1969 from the high rate of $ 2.8 billion in 1968 when demand pressures during a great part of the year were still moderate. But exports had been inflated in the last quarter of 1968 by special factors which shifted perhaps to $ 0.3 to 0.4 billion from 1969. Adjusting for this, the swing was therefore small;

Table 6 **Balance of Payments**
$ million

	1968	1969	1968		1969	
			I	II	I	II
Seasonally adjusted:						
1 Exports, fob	24 878	29 119	11 715	13 163	13 765	15 354
2 Imports, fob	19 202	23 986	9 173	10 029	11 488	12 498
3 Trade balance	5 676	5 133	2 542	3 134	2 276	2 851
4 Services and private transfers, net	–1 785	–2 199	–737	–1 048	–974	–1 225
5 Official transfers, net	–1 035	–1 129	–472	–563	–554	–575
6 Current balance	2 855	1 805	1 333	1 522	748	1 057
Not seasonally adjusted:						
7 Current balance	2 855	1 805	1 252	1 603	874	931
8 Long-term capital[1]	–2 908	–5 928	–1 152	–1 756	–2 440	–3 488
8.1 Private	–2 539	–5 486	–1 025	–1 514	–2 419	–3 067
8.2 Official	–369	–442	–127	–242	–21	–421
9 Basic balance	–53	–4 123	100	–153	–1 566	–2 557
10 Short-term capital and unrecorded[2]	1 197	23	681	516	2 840	–2 817
11 Balance on non-monetary transactions	1 144	–4 100	781	363	1 274	–5 374
12 Commercial banks, net	614	1 185	–61	675	–1 090	2 275
13 Balance on official settlements	1 758	–2 915	720	1 038	184	–3 099

1 Excluding special transactions.
2 Including advance payments for military imports.
NOTE Details may not add, due to rounding.
Sources: Deutsche Bundesbank, OECD.

in the second half of 1969 the surplus was running at an annual rate of $ 2.1 billion. The decline in the " underlying " surplus was mainly due to invisibles. The seasonally adjusted trade balance, taking account of the shift from 1969 to 1968 referred to above, actually improved. This development was in marked contrast to the 1965 experience when strong demand pressures at home was accompanied by a marked swing of Germany's current account into a deficit position. The situation in 1969 was influenced by the fact that demand pressures were strong in many other countries also. Nevertheless, there are indications that pressures were relatively stronger in Germany than elsewhere. The persistence of a large current account surplus in these conditions provides a measure of the disequilibrium that had developed.

With the unexpectedly fast expansion of world trade, Germany experienced another year of very strong export growth (see Table 7). This was mainly attributable to a further large increase in investment goods exports, where Germany has a traditionally strong market position. However, in spite of growing pressures on resources, Germany not only maintained her overall relative competitive position but even managed to increase exports about 3 per cent more rapidly than the growth of markets, which is well above the medium-term trend. It should be noted that the improvement in the relative position occurred largely during the second half of 1969. The first half of 1969, when German unit values rose at rates similar to those of competitors, saw little change in German market shares[1]. But in the second half Germany not only maintained her previous half year share in growth of total OECD exports to 15 geographical markets, but increased it (in dollar terms) by some 2 percentage points. During this period, mostly in consequence of the 9,3 per cent revaluation in the autumn, dollar unit values of German exports rose by almost 3 per cent more than those of her main competitors. This meant that in volume terms the German export performance deteriorated somewhat. In judging German export performance during 1969 it has also to be taken into account that during the first 9 months of the year German exports were inflated by anticipatory forward purchases due to the expected DM revaluation.

Imports responded strongly to the rapid expansion of aggregate demand and the import/expenditure elasticity reached its highest level over the last decade (Table 8). The acceleration in the growth rate of total merchandise imports was mostly attributable to a very fast rise in investment goods imports (see Table 7). This, in turn, reflected the very strong investment boom in Germany.

1 The measurement takes into account the speculative shifts.

Table 7 **Foreign Trade**

	Exports fob				Imports cif			
	Weights in 1969 total	1967	1968	1969	Weights in 1969 total	1967	1968	1969
		Percentage value change from previous year				*Percentage value change from previous year*		
Products of agriculture, forestry, and fisheries	1.1	11.6	13.1	27.2	15.1	–7.4	1.1	11.3
Mining products	2.1	–3.3	11.1	–4.7	9.6	7.8	18.8	6.7
Basic materials and semi-finished goods	27.3	12.1	12.2	11.5	31.0	0.3	26.3	23.9
Manufactured foods and tobacco	2.5	26.9	20.8	17.1	8.1	–4.2	10.9	13.6
Other consumer manufactures	11.2	7.7	16.1	18.8	13.6	–14.5	25.4	25.3
Investment goods	54.3	5.9	15.3	15.1	18.9	–1.9	17.0	31.6
Other goods	1.5	5.3	8.5	9.0	3.7	–2.8	–15.7	27.5
Total	100.0	8.0	14.4	14.1	100.0	–3.4	15.7	20.7
		DM billion				*DM billion*		
Total merchandise exports/imports		87.0	99.6	113.6		70.2	81.2	98.0

Source: Statistisches Bundesamt.

Table 8 **Import/Expenditure Elasticities**[1]

1959	1960	1961	1962	1963	1964	1965	1966	1976	1968	1969
1.6	1.5	0.7	1.3	1.2	1.2	1.8	0.6	–10.3	1.7	1.9

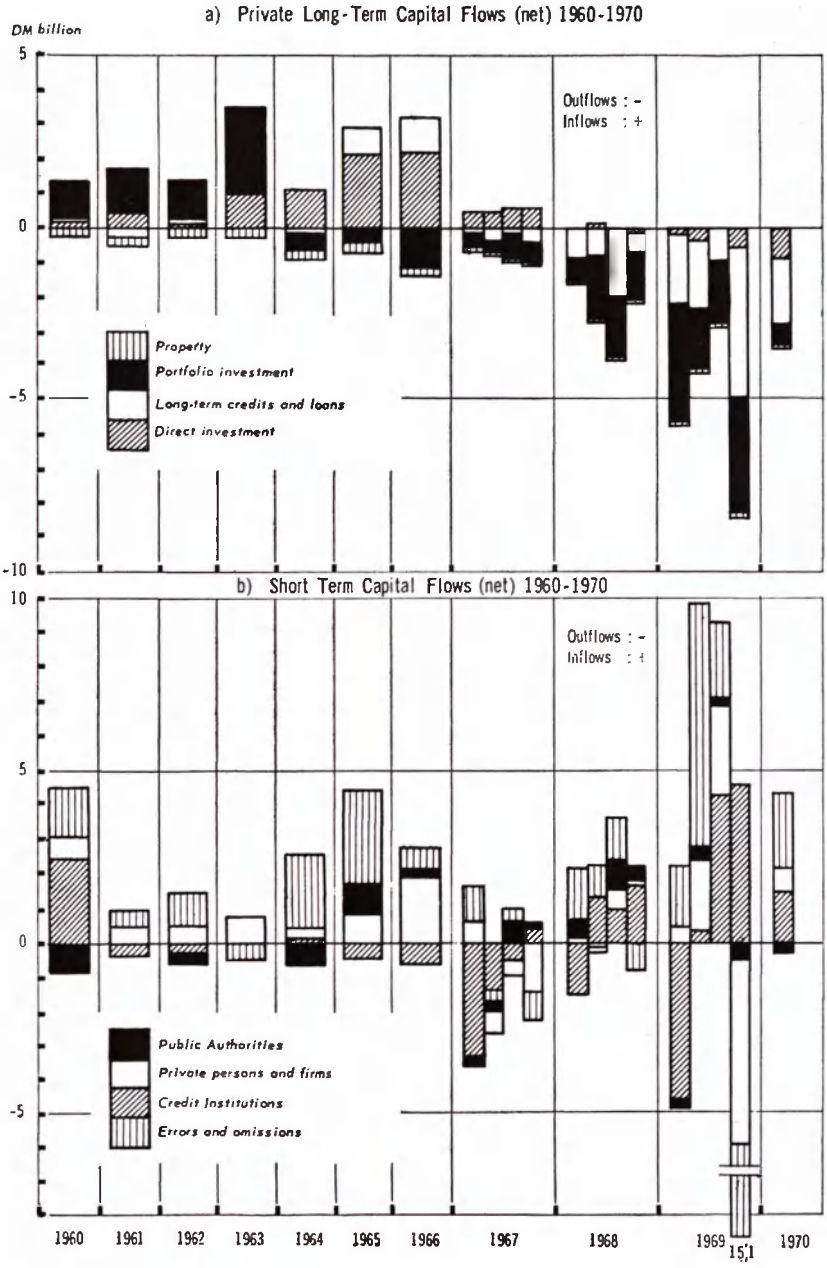
1 Percentage increase of total merchandise imports ($ value) divided by percentage increase of tota aggregate demand.

With an increase by about $ 200 million the deficit on services rose only slightly. This deterioration can be largely explained by increased foreign travel expenditure. With the strong growth of personal incomes, travel expenditure abroad, which had remained constant during 1966-68, recovered sharply. But net expenditure on commissions, advertising, patents and licences also rose strongly. Part of this deterioration was offset by a fall in the deficit on the balance of investment income, reflecting the considerable increase in interest and dividend receipts related to the extraordinarily strong short-and long-term capital export of recent years and an increase in the return on Bundesbank assets invested abroad. The deficit on transfers was rather higher than in 1968, largely due to sharply increased remittances of foreign workers. However, the net outcome on the invisibles account was probably distorted by speculative movements prior to the DM revaluation[1].

The long-term capital outflow rose sharply to some $ 5.9 billion in 1969, entailing a deficit in the basic balance of some $ 4 billion. The extraordinarily large outflow of long-term capital, corresponding to almost 60 per cent of savings by private households, meant that Germany became the world's biggest exporter of long-term funds. An outflow of this size, almost wholly on private account, was unprecedented. Over the whole decade of the 1950's the outflow had scarcely totalled $ 0.5 billion. From 1960 to 1966 Germany was even a net importer of private long-term capital at an average annual rate of some $ 0.4 billion. The first sizeable net private capital outflow occurred during 1968. In that year, due to a sharp rise in portfolio investment and in long-term credit the long-term outflow amounted to $ 2.5 billion.

1 Invisible transactions are, as a rule, recorded on a settlements and not a transactions basis. For this reason all invisible items are subject to leads and lags which develop in periods of monetary unrest. Significant examples are foreign travel receipts and expenditures: in principle, the main statistical basis of this balance are movements of banknotes. Banknotes denominated in a currency subject to revaluation speculation are kept rather than sent back to the country of origin, thus entailing underrecording of that country's travel expenditure. Flows of investment income are also liable to be significantly affected by revaluation speculation.

Diagram 6 Capital Flows



Source: **Deutsche Bundesbank.**

Most of the long-term capital outflow in 1969, some $ 5½ billion, was on private account, of which the banks accounted for roughly half. Increasingly restrictive monetary policies abroad and the comparatively low level of German interest rates led to a trebling of the banks' long-term lending abroad; this lending accounted for almost 84 per cent of their total long-term capital exports (in 1968: 64 per cent). The composition of long-term capital exports of the non-bank sector also changed considerably. Almost two thirds consisted of purchases of foreign shares and investment certificates and of DM denominated foreign bonds issued in the German capital market. The total outflow on non-bank portfolio account was even higher (some $ 2 billion) which was mostly due to the fact that after the revaluation foreigners liquidated some $ ¼ billion of German securities. There was finally an important swing in direct investment. Following the previous year's trend, and mostly due to the US balance of payments programme inducing US companies and their foreign subsidiaries to finance direct investment abroad by funds raised in foreign markets, there was a further fall in inflows associated with foreign direct investment in Germany, whereas German direct investment abroad rose. For the first time since World War II the direct investment account of private non-banks showed a significant net outflow ($ 0.2 billion). The long-term capital outflow

Table 9 **Long-Term Capital Transactions**
Millions of DM (net capital exports:—)

	1966	1967	1968	1969
A Banks	419	–1 107	–5 156	–10 784
Of which:				
Advances and loans	(297)	(–887)	(–3 299)	(–9 104)
Foreign DM bonds	(15)	(–81)	(–1 468)	(–1 044)
B Private non-banks	1 350	–583	–5 001	–10 352
Direct investment	2 137	1 771	19	–894
Portfolio investment	–1 201	–1 796	–3 887	–8 856
Of which:				
Foreign DM bonds	(–520)	(–42)	(–2 228)	(–3 171)
Other bonds	(–82)	(–240)	(–251)	(–798)
Foreign shares	(–111)	(–598)	(–473)	(–1 573)
Foreign investment fund certificates	(–152)	(–297)	(–948)	(–2 106)
German securities	(–336)	(–619)	(+13)	(–1 208)
Advances and loans	735	–259	–774	–140
Other	–321	–299	–359	–462
C Official	–2 531	–1 513	–1 366	–1 839
D Total long-term capital transactions	–762	–3 203	–11 523	–22 975

Source: Deutsche Bundesbank.

has remained high since the DM-revaluation; and indeed in the fourth quarter of 1969 it attained a record level of $ 2.6 billion. In January 1970 long-term capital exports continued on a fairly high scale ($ 0.6 billion) again roughly half on the account of the banks' almost completely in the form of loans and credits. The banks' continuation of the very high capital exports during a period of increasingly severe liquidity tightness was due to the fulfilment of commitments made earlier when liquidity was ample. Recent evidence suggests, however, that the banks have refrained since the end of 1969 from increasing their credit commitments abroad. Between the middle of December and the end of February issues of foreign-DM bonds ceased completely.

There were several reasons for the very high level of long-term capital exports in 1968 and 1969:

(*a*) For the first time in the post World War II period domestic long-term rates were much lower than those in other principal markets. There was notably a persistant interest rate differential between Euro-dollar and Euro-DM bonds which narrowed only recently under the pressure of the growing liquidity drain (See Diagram 8). Until the revaluation the cost advantage from long-term borrowing in Germany apparently more than compensated the loss expected from revaluation of the Deutschemark;

(*b*) Access to the German capital market is completely free while quantitative restrictions on capital outflows were introduced or extended in many other countries;

(*c*) Over most of the period foreign-DM bonds offered better yields than competing domestic issues (see Diagram 9). The acquisition of such bonds was, therefore, attractive for German residents;

(*d*) Until September 1969, the banks' heavy engagement in long-term investment abroad was related to their favourable liquidity situation which was connected with a continued strong inflow of medium-and long-term deposits emanating from the enterprise sector. In 1968 this inflow reflected the sharp increase of the internal cash flow of this sector. During the first 9 months of 1969 a very important part of the large increase in medium-term deposits resulted indirectly from inflows of funds from abroad in anticipation of DM-revaluation, and was thus highly volatile.

Revaluation had a strong effect on the short-term capital balance. During the first nine months of the year the inflow of short-term funds was well in excess of the large basic deficit ($ 2.0 billion) so that the balance of non-monetary transactions showed a surplus of some $ 2.1 billion. This

implied that private non-banks experienced a speculative short-term inflow of some $ 4 billion (see Table 10) which was completely reversed during the last quarter of 1969. Thus, during the year as a whole, most of the basic deficit was reflected in a fall in official reserves, which, at the end of 1969, were about $ 3 billion lower than a year earlier.

Table 10 **Speculative short-term capital movements before and after DM revaluation**
$ billion

	1969	
	1st-3rd quarter	4th quarter
1 Total enterprise sector	4.0	–4.0
of which:		
Short-term credits	1.3	–1.5
Errors and omissions	2.7	–2.5
2 Credit institutions		
Short-term liabilities vis-a-vis foreign banks[1]	0.7	–0.2
Total	4.7	–4.2

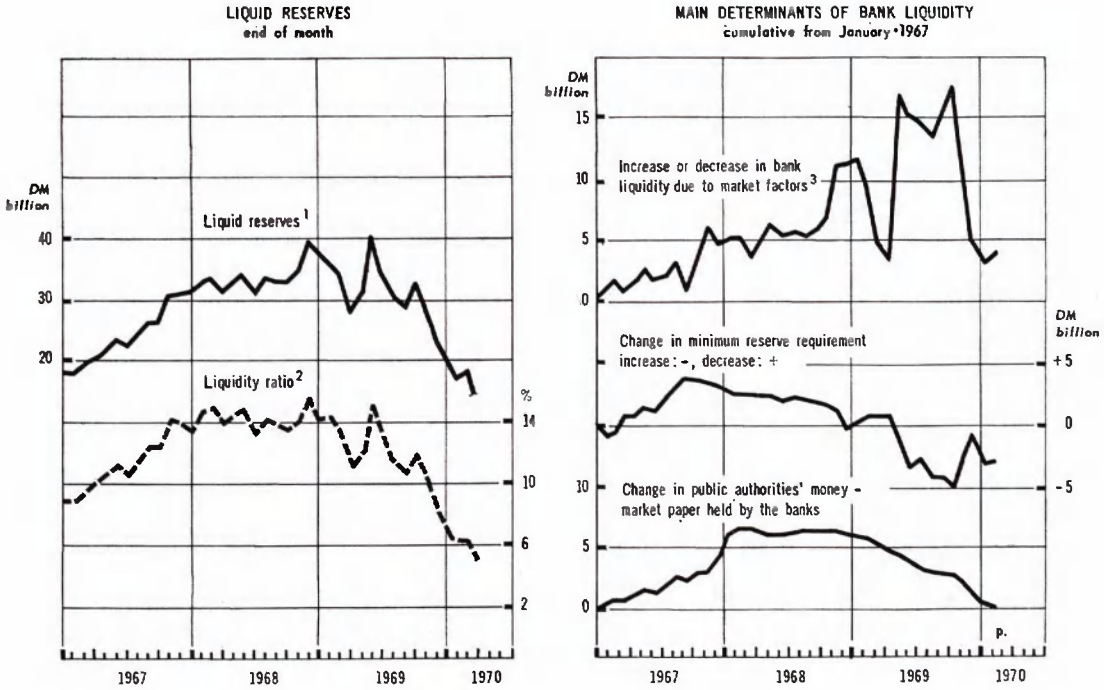
1 Excluding December 1969.
Source: Deutsche Bundesbank.

II MONETARY AND FISCAL DEVELOPMENTS

Monetary Policy

Monetary expansion continued at a high rate throughout 1969. On domestic grounds the German authorities would have preferred tighter monetary conditions before the revaluation, but the massive speculative short-term capital inflows kept the banking system highly liquid. Nevertheless, beginning in March the Bundesbank was active in using the existing room for restrictive monetary policy measures, notably through changes in minimum reserve requirements (see Annex 1). Another major restrictive impact on the bank liquidity resulted from public finance operations (see chapter on fiscal policy). Furthermore, as already noted, until the revaluation of the Deutschemark the Bank succeeded in maintaining a sufficiently high interest rate differential towards the Euro-Dollar Market to encourage a high level of long-term capital exports. These developments led to a fall in the banks' liquidity ratio from 14½ per cent of total deposits at the end of January 1969 to about 12 per cent in September 1969. This was the more remarkable as during the same period Germany experienced a hot money inflow of almost DM 20 billion.

Diagram 7 Bank Liquidity



1 Central bank balances, domestic money-market paper, money-market investment abroad and unutilised rediscount quotas, less minimum reserve requirements and Bundesbank advances against securities.
2 Ratio of liquid reserves to deposits in percentage.
3 Excluding change in money-market debt shown below.
p Provisional.

Source: Deutsche Bundesbank.

The establishment of the new DM-parity changed the monetary situation fundamentally. By the end of January 1970 the sharp fall in official reserves had led to a reduction in bank liquidity of some DM 15½ billion, offset only to a small extent by reductions in minimum reserve requirements. At the end of January the liquidity ratio of the banking system amounted to some 6 per cent of total deposits. This corresponded roughly to the lowest ratio reached during the tight money period of 1966. By the end of March the ratio had fallen further to about 5¼ per cent.

Although domestic short-term rates reacted promptly to the September-increases in the discount and Lombard rates, the sharp reduction and the final disappearance of the forward premium on DM after the revaluation increased the effective (covered) spread between short-term rates in Germany and those obtaining in the Euro-Dollar market in favour of the latter. During October/November the outflows of funds from Germany, therefore, included substantial amounts of banking funds taking advantage of interest rate differentials. The banks drew heavily on their refinancing possibilities with the Bundesbank, especially on Lombard credits. The increase in the Lombard rate[1] to 9 per cent in early December, and to 9½ per cent in March produced a corresponding upward movement of domestic short-term rates, reduced the interest rate differential in favour of Euro-Dollar rates. By the end of January, the incentives for short-term money exports had virtually disappeared (see Diagram 8). This new situation and the increasing degree of monetary tightness induced the banks to repatriate certain amounts of funds. During the first quarter of 1970 repatriation and short-term borrowing abroad attained a level of almost DM 1 billion.

The expansion of bank credit accelerated strongly during 1969, notably during the final quarter when the reversal of speculative funds invested with German firms led to a substitution of foreign by domestic credit. Since then credit expansion has shown first signs of flagging. The expansion of the volume of money has, however, continued. Total private sector borrowing showed an increase in the course of 1969 which was some DM 20 billion higher than during 1968. However, with the increase in pubic borrowing diminishing sharply, total domestic expansion of bank credit (including bank purchases of non-bank bonds) grew only by some DM 10¾ billion more than during 1968. The small increase in total loans to public authorities, the lowest since 1963, reflected the unexpectedly rapid growth of tax revenue during the year.

1 The Lombard rate is the rate on short-term bank loans against eligible securities deposited as collateral with the Central Bank. In a situation where discount quotas are exhausted, resort to Lombard credit becomes the marginal source of refinancing. This higher rate, then, automatically determines money market rates.

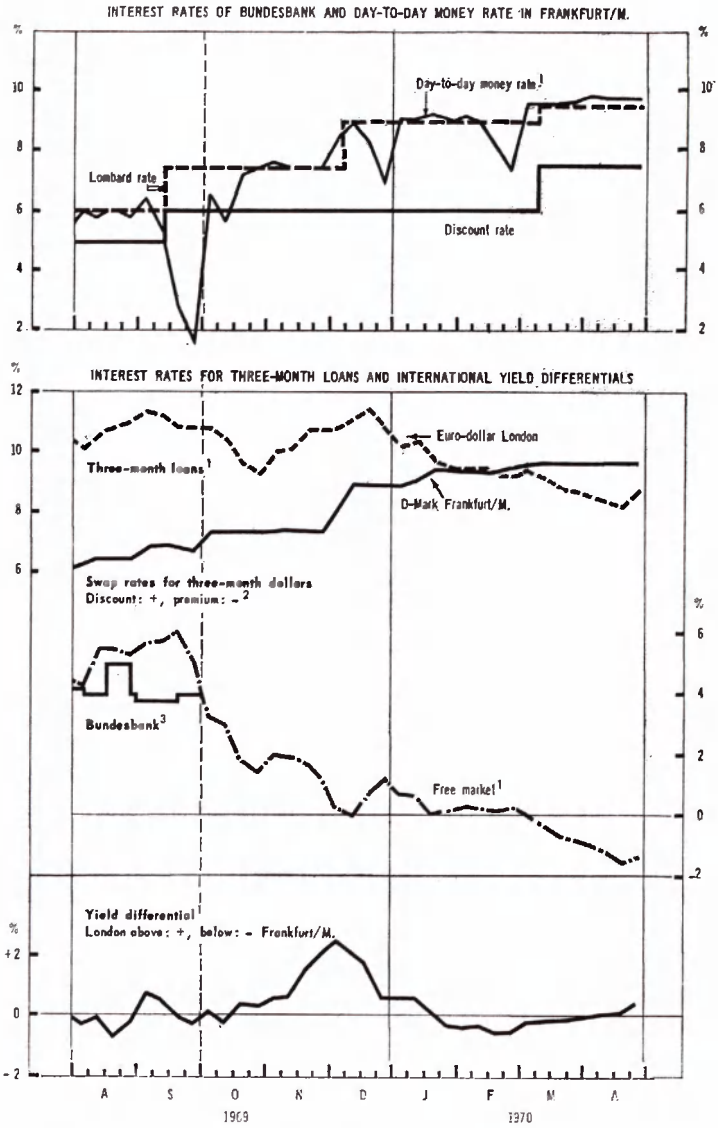
" Monetary capital formation "[1] although high, was some DM 4 billion lower than the record level attained in 1968. The decline was fully attributable to developments after the revaluation which, in the last quarter, entailed a DM 5 billion fall in medium-term bank deposits[2]. The sharp slowdown in monetary capital formation during that quarter was accompanied by an important increase in the banks' domestic and foreign long-term lending. This grew by DM 7 billion more than a year earlier, thus almost fully accounting for the total additional increase in 1969 (DM 8 billion). The fast expansion of total long-term lending during a period of a sharply diminishing medium-and long-term deposit inflows reflected the fact, already noted, that the banks had to meet credit commitments made earlier. Thus it appears that the banks do not seem to have been fully aware of the vulnerability of their liquidity position. This was due to the fact that an important part of the large increase in medium-term deposits until September 1969 resulted indirectly from revaluation speculation and was thus highly volatile. The fast reversal of speculative flows during the final quarter of the year was, therefore, not only related to a corresponding liquidity loss but also with a sharp deterioration in the maturity structure of bank liabilities. To this extent the increased transformation activity of the banking system during 1969 was involuntary.

Restrictive central bank measures together with increasingly restrictive monetary policies abroad resulted in a general upward pressure on interest rates, reinforced after the revaluation by the disappearance of the large forward premium on the Deutschemark. Towards the end of the first quarter of 1970 the cost of overdraft credits to non-banks had risen to about 11-12 per cent, i.e. almost 4 percentage points higher than one year earlier. During the same period the average bond yield rose by about 1½ percentage points to 8 per cent and Euro-DM yields rose by about 2 percentage points to 8¾ per cent, which is roughly the rate for Euro-Dollar issues. By the first quarter, then, the cost-advantage of borrowing in the Euro-DM market had virtually disappeared. With liquidity increasingly tight, the banks progressively reduced their bond market participation rate, which had been abnormally high since the beginning of 1967. By the end of 1969 the banks' purchases in the bond market had come to a standstill.

Net domestic issue activity on the bond market, which had attained a post-war record level in 1968, dropped by about DM 5¼ billion in 1969. The decline was mainly due to a sharp fall in the volume of public issues reflecting voluntary restraint on new issue activity and the rapid growth of tax revenue far above the original budget estimates. There was, however,

1 Savings deposits, bank bonds held by non-banks, time deposits exceeding 3 months and increases in capital and net published reserves of banks.
2 Time deposits with maturities of 3 months up to 4 years.

Diagram 8 Money-Market Rates and International Yield Differentials



1 Average of weekly bank-return dates.

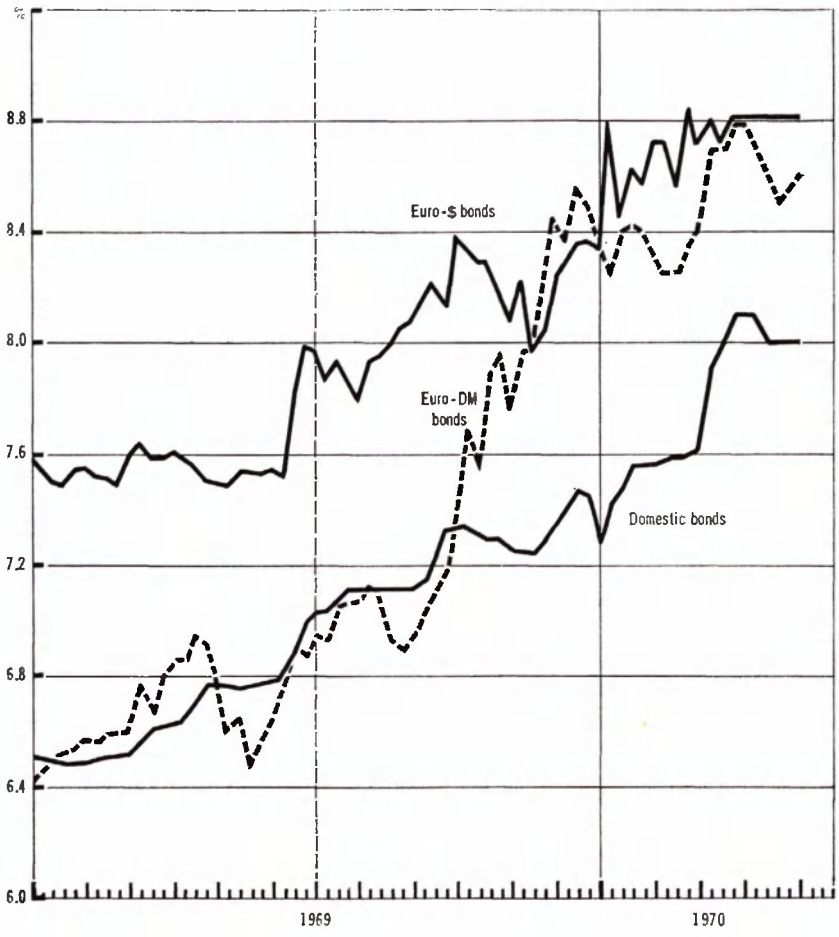
2 " Discount " = deduction to be accepted as cost of forward exchange rate cover on selling forward dollars,

" premium " = yield obtainable on selling forward dollars.

3 Swap transactions suspended since September 29.

Source: Deutsche Bundesbank.

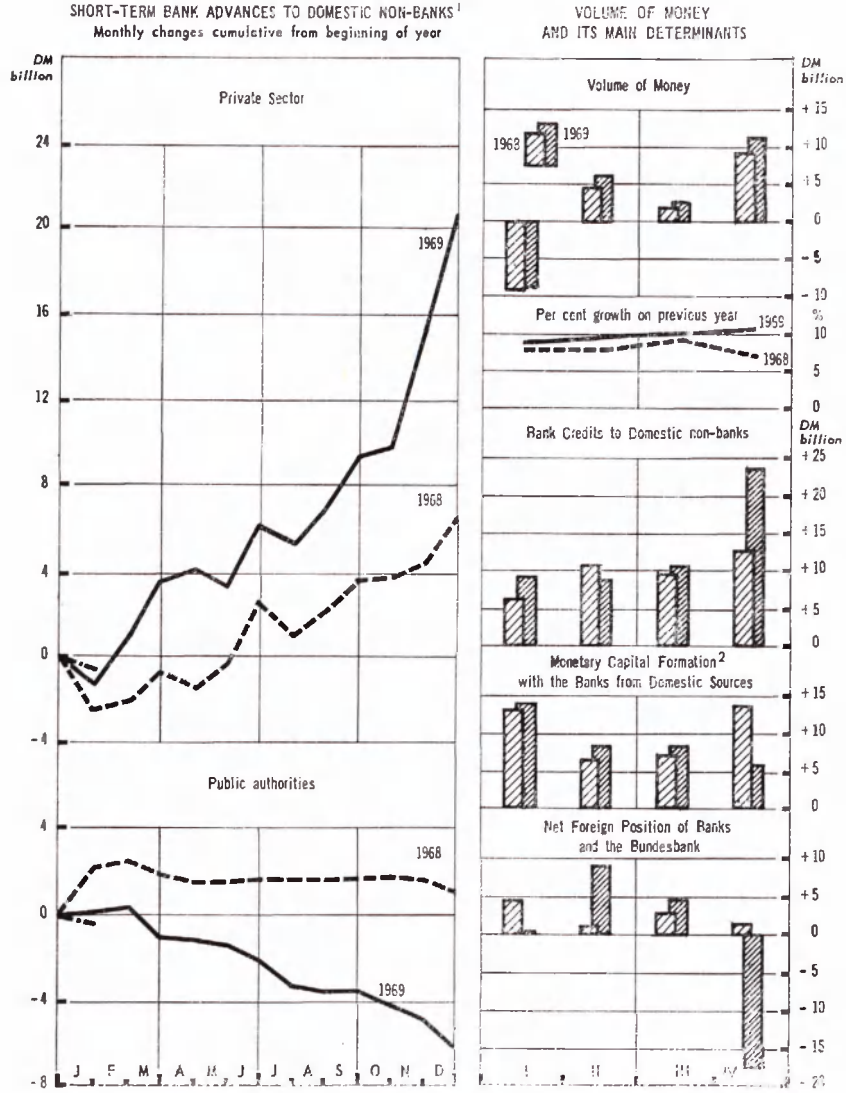
Diagram 9 Bond Yields



Source: Deutsche Bundesbank.

a further important increase in issues of DM denominated Euro-bonds to DM 5½ billion, representing almost 40 per cent of all foreign issues floated in Europe in 1969. Some 80 per cent of the DM-issues were placed with domestic purchasers. The bond market therefore contributed decisively towards increasing private capital exports. Domestic purchases of foreign bonds, together with the big net outflow due to large purchases of foreign shares and investment certificates, represented an outflow on portfolio

Diagram 10 Credits, Volume of Money and Monetary Capital Formation



1 Excluding Bundesbank.

2 Savings deposits, time deposits exceeding 3 months and increases in capital and net published reserves of the banks.

account equivalent to some 80 per cent of total rise in domestic net issues in bond and share markets (compared with some 30 per cent a year earlier). As during previous periods of extreme liquidity tightness the quantitative effects of monetary policy were most marked in the bond market. During the first quarter of 1970 total net issues of fixed interest securities fell to less than half the year-earlier level. The banks were probably net sellers of bonds during this period. There was also a growing disinclination on the part of other domestic investors to invest in bonds. To avoid overstraining the market, the Central Capital Market Committee pursued a cautious policy; new issue activity was reduced to a minimum. At the beginning of March it was decided to end the new issue pause which had been practically complete during December and January and the market was successfully tested with some representative issues bearing coupons of 8 per cent and yielding up to 8½ per cent.

Table 11 **Bond Market - New Issues**[1]
DM billion

	1966	1967	1968	1969
Net issue of domestic securities	5.0	15.0	17.7	12.4
Public authorities	0.6	4.9	3.8	0.3
Credit institutions	4.4	9.2	13.9	12.3
Other private	—	0.9	–0.1	–0.2
Net increase in domestic holdings of foreign bonds	0.6	0.5	4.1	5.3
Net bond purchases:				
Central Bank	—	1.3	–0.4	–0.5
Credit institutions	1.6	11.7	14.6	9.8
Domestic non-banks	4.1	3.3	7.7	9.4
Foreign investors	–0.1	–0.8	–0.2	–1.0
Memorandum items:				
Total capital export via bond market	0.7	1.3	4.3	6.3
Total capital export via equity market	—	0.7	2.0	5.4
A Total capital export via security markets	0.7	2.0	6.2	11.7
B Total domestic issues in bond and equity markets	7.7	16.9	20.8	15.2
C A in per cent of B	9.1	11.8	29.8	77.0

1 Net of repayment and repurchases at issue value. Detail may not add due to rounding.
Source: Deutsche Bundesbank.

Fiscal Policy

As late as December 1968—following the introduction of the border tax measures—the Financial Planning Council had recommended moderate expansionary measures in order to compensate for the expected deflationary impact of the quasi-revaluation, but during 1969 the posture of fiscal policy

became progressively more restrictive. In early March, when economic indicators for the first months of the year suggested the risk of an excessive rise in demand, the authorities proposed or adopted certain restrictive budgetary measures and these were reinforced before mid-May. The most important of these measures can be summarised as follows :

(*a*) 1969 expenditure plans of the Federal Government were reduced by about DM 1.8 billion, mainly affecting public consumption and investment;

(*b*) The Financial Planning Council recommended that the Federal Government and the Länder set up a contracyclical reserve of DM 3.6 billion (in accordance with para. 15 of the Growth and Stability Law). This was approved by Parliament in June. It was, however, left to the authorities' discretion whether they would use these funds for redemption of money market paper;

(*c*) It was also decided that any Federal Government or Länder tax revenue resulting from a more rapid growth of GNP than that assumed in the original budget calculations and not needed to meet the above requirements, should either be used for increased redemption of money market paper or put into the contracyclical reserve;

(*d*) Corporate and income tax payments, usually lagging considerably behind the accrual of income during the upswing, were speeded up by an increase in tho size of prepayments. According to a Council of Economic Advisers estimate, this increased tax revenue by about DM 1.5 billion in 1969.

The economic effects of utilizing a contracyclical reserve for redemption of money market paper held by the banking system have been misunderstood by some German observers. According to their view[1] these operations do not contribute to dampening economic activity as they cancel the restrictive liquidity impact of the accumulation of the contracyclical reserve. However, this line of argument ignores the fact that in the first instance the monetary base bas been reduced by the amount of the contracyclical reserve. As far as bank liquidity is concerned it would not appear to make any difference whether the funds are neutralized with the Central Bank or whether the Government utilizes the contracyclical reserve to redeem short-term money market paper placed with the banking system. In both cases the original restrictive impact of reducing the monetary

1 See for example Jahresgutachten 1969-70 des Sachverständigenrates zur Begutachtung der gesamtwirtschaftlichen Entwicklung. p. 47.

base is maintained, and the banking system loses the corresponding amount of reserves. It is true that for the banking system short-term debt redemption represents an exchange of secondary liquidity against primary liquidity. But at the end of the operation the banking system is not more liquid than before. Secondary reserves can be exchanged for primary reserves at any time through normal transactions with the Central Bank. Finding themselves with too high a primary liquidity ratio (for reasons of profitability the banks tend to keep their primary liquidity ratio low) the banks will use their excess primary liquidity to restore their initial secondary liquidity ratio by buying money market paper.

Fiscal policy did not exert a restrictive impact on bank liquidity only by the accumulation of a contracyclical reserve. Public authorities also carried out important debt management operations, refunding short-term money market paper, taking up bank credit or issuing medium-and long-term bonds. During 1969 for example the amount of outstanding Federal treasury bills was reduced by some DM 6 billion and the amount of outstanding long-term bank credits increased by almost DM 4½ billion. These operations contributed decisively towards the reduction of bank liquidity in 1969. This contrasts with the situation in 1968 when public sector debt management operations had increased bank liquidity by some DM 2 billion. This swing in debt management policy provided an effective complement to restrictive monetary policy measures. (See Diagram 7).

The restrictive impact of public finance is also apparent from a closer analysis of the General Government account. Mainly as a result of the strong upswing of the economy, total receipts grew by some 16 per cent. This was far above the original estimates which were based on a current revenue rise of some 10 per cent. In contrast, with the effect of expenditure cuts cancelled by large and unexpected wage increases in the final quarter, the original budget estimate of an expenditure rise of some 9½ per cent corresponded closely to the actual outturn. This meant that there was an important swing from a financial deficit of some DM 4 billion in 1968 to a surplus of DM 9 billion in 1969. Only part of the swing resulted from discretionary policy measures; a large part reflected the impact of automatic stabilizers on the revenue side.

An estimate of the demand impact of the changes in the General Government Budget points to a deflationary impact of some 2 per cent of GNP in 1969.

The 1970 budget envisages a continuation of restrictive fiscal policy. The 1970 developments shown in the table above reflect the following measures and recommendations:

(*a*) The expenditure rise in the draft Federal budget (cash basis) has been limited to 8¾ per cent. This was made possible by a temporary blocking of DM 2.7 billion to be released only if economic activity shows signs of flagging in the course of the year. Roughly half the temporary cuts affect Federal expenditure for investment purposes, mostly in the construction sector;

(*b*) According to the guidelines adopted for the execution of the budget, public spending should as far as possible be postponed from the first to the second half of 1970;

(*c*) The Federal Government has decided that the DM 0.7 billion tax cut for wage and salary earners, originally planned for the beginning of 1970, should be postponed until July 1st 1970[1]. The first stage in abolition of the 3 per cent supplementary levy on income and corporate taxes will be postponed until the beginning of 1971. Important new savings incentive measures (the increase in the limit on savings deductable from taxable income from 312 DM to 624 DM) will be introduced already in 1970;

(*d*) The Financial Planning Council has agreed that the Federal Government and the Länder set up a contracyclical reserve of DM 2.5 billion. The corresponding law has been promulgated in accordance with para. 15 of the Growth and Stability Law. The funds (1.5 billion Federal Government, 1.0 billion Länder) have to be constituted with the Central Bank by the end of June. In contrast to 1969 no provisions have been made according to which debt could be redeemed as an alternative to setting op the reserve;

(*e*) The Financial Planning Council has furthermore agreed on the following recommendations.

1 The Länder should try to block temporarily some DM 1-2 billion of expenditure to be released only when the economic situation renders this appropriate. The Communes should also endeavour to block expenditure;

2 In addition to the above measures, Länder and Communes should try to delay the placing of additional orders to the second half-year.

Some of these measures have already been adopted.

If realised in full, the above measures should result in a further rise in the General Budget financial surplus from DM 9 billion in 1969 to some

1 For technical reasons it would, however, appear unlikely that the tax cut will affect domestic demand before Jan. 1st 1971.

Table 12 **The Budget of the Total Public Sector - Estimated Impact on Domestic Demand**
DM Billion and Per cent

General Government	1968	1969[1]		1970[2]	
	DM Bill.	DM Bill.	Percentage increase	DM Bill.	Percentage increase
Receipts, total	212.2	245.9	16	272	10½
of which: direct taxes paid by households	105.0	116.3	11	136	17
Expenditure, total	216.1	236.8	9½	257.5	8¾
Public consumption	83.6	93.2	11½	103	10½
Transfers, etc.	86.6	91.9	6	99	7½
of which: transfers to private households	74.2	79.6	7½	84¾	6½
Financial savings	–4.0	9.1		14½	
Estimated impact of Budget changes on domestic demand (% of GNP)[3]		–2		–½	

1 Official estimates for fiscal years.
2 Secretariat forecasts.
3 The Secretariat estimates of the economic impact of the Budget refer to all changes in budgetary items and, thus, include built-in stabilisers as well as discretionary measures. Secondary effects on personal income and private consumption are taken into account.

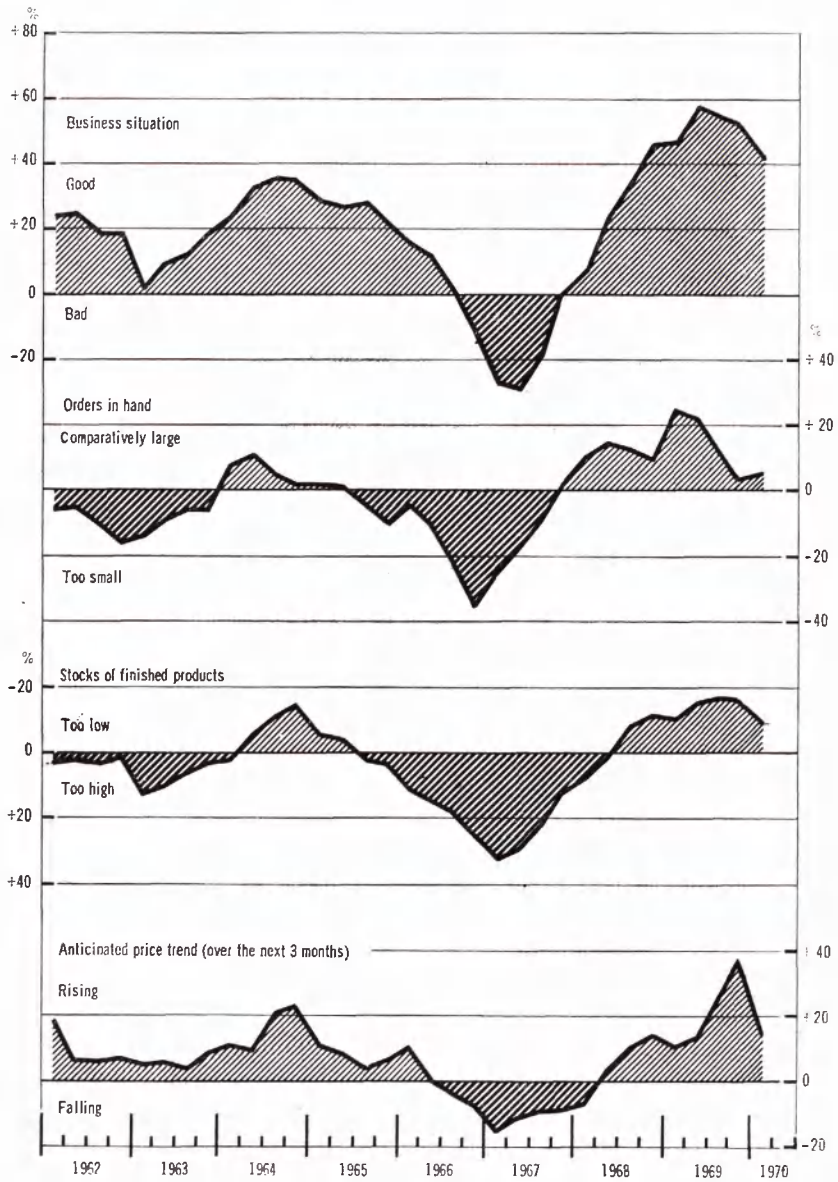
DM 15 billion. Taking into account the changes in expenditure and revenue items, the Budget might exert a deflationary impact equivalent to ½ per cent of GNP. During the first half of the year expenditure on goods and services should rise relatively little on a seasonally-adjusted basis. It must also be taken into account that—as during 1969—the Budget is likely to exert a considerable restrictive impact on bank liquidity.

III PROSPECTS AND POLICY ISSUES

Recent indicators suggest that the expansion of activity has continued at a high rate in the early months of 1970, though unusually bad weather hampered production in some sectors, notably construction, in January and February. The rise in industrial production remained strong in the first quarter, with the utilisation of industrial capacity at or exceeding the record rate of 96 per cent[1] attained in the last quarter of 1969. The labour market

1 Estimate by Deutsches Institut für Wirtschaftsforschung.

Diagram 11 The Business Situation in Industry[1]



1 Percentage balances from the replies of entrepreneurs. From 1966 change in coverage of survey.

Source: IFO.

has remained tight, with seasonally adjusted unemployment still at only 0.7 per cent in March. Between January and March the number of foreign workers rose by around 125 000 to the record figure of 1.7 million. Nevertheless, the number of vacancies for foreigners was still as high as 100 000. The total number of vacancies in March amounted to 890 000 (710 000 a year earlier) against 155 000 unemployed (seasonally adjusted).

Recent surveys suggest that, in spite of the revaluation and the marked tightening of monetary conditions, business expectations remain relatively buoyant. The order backlog is still growing, but the smaller volume inflow of new orders since revaluation, particularly export orders, and tighter demand management policies, point to a less uniformly strong demand situation in coming months. With supply constraints still acute in some sectors a return to more normal rates of capacity utilisation can be expected to develop only gradually.

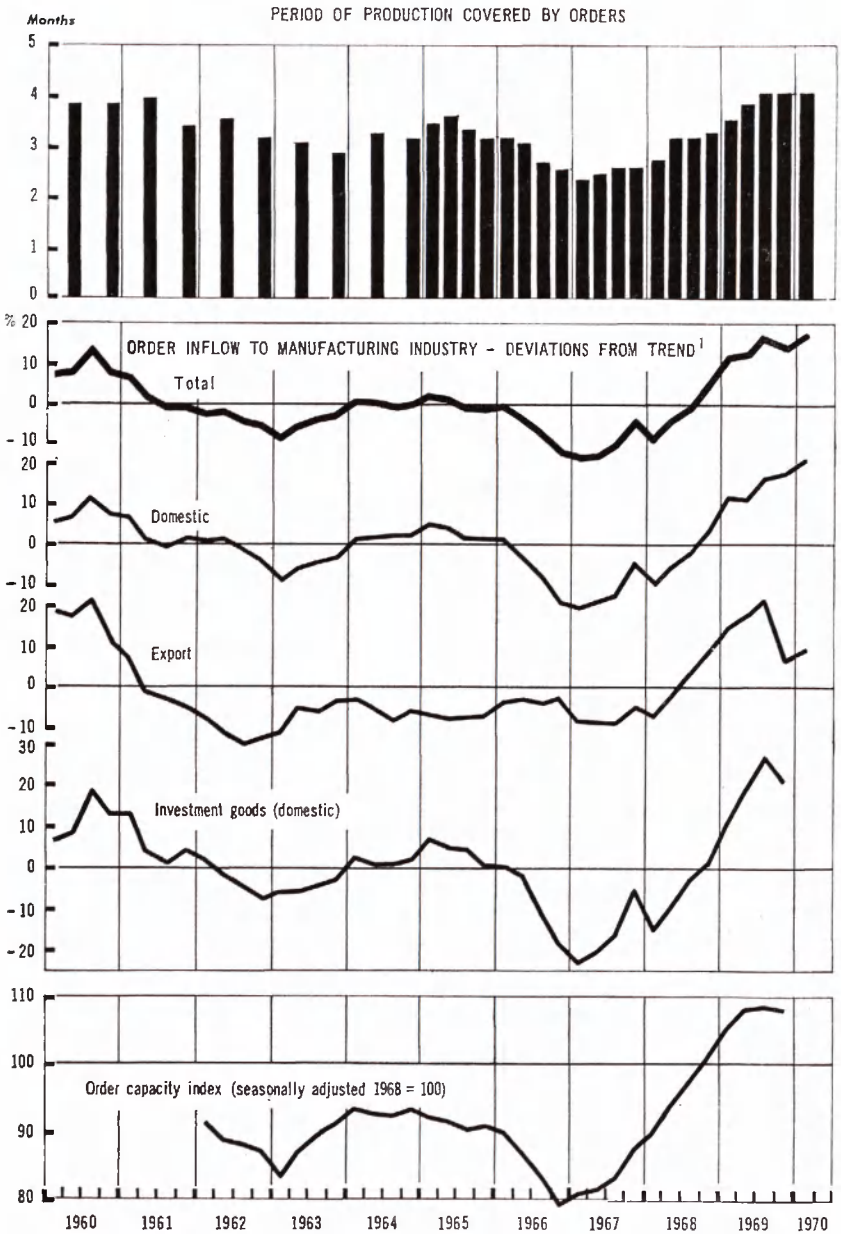
The present assessment is made on the assumption that existing policies are maintained unchanged. The fiscal policy adopted at the beginning of the year entailed no major changes, compared with last year, in tax rates or in the rate of growth in real terms of public purchases of goods and services. It is assumed that, in accordance with the guidelines adopted for the execution of the budget, public spending will as far as possible be postponed to the second half of 1970. In the first half of 1971, it has been assumed that, apart from the tax cut mentioned on page 34, tax rates will remain unchanged, and that public purchases of goods and services will grow at the same rate as between 1969 and 1970. Monetary policy is assumed to remain tight. The present view of the Bundesbank with regard to the required aims of demand management policy is " that the expansion of demand must be curbed not merely to the extent that the margin for real growth narrows but by substantially more "[1].

Internal Prospects

In the first half of 1970, the expansion of home demand would seem likely to remain strong, led by a further sharp increase in business fixed investment. The IFO survey of investment intentions taken in November pointed to a 16 per cent rise in the value of fixed investment in manufacturing between 1969 and 1970. On the basis of price expectations at that time, this may have implied a volume rise of some 10 per cent. Most of the increase seems likely to take place in the first half-year, given the large volume of projects in the pipeline. At the beginning of the year the order backlog of the investment goods industries amounted to the equivalent of

1 Deutsche Bundesbank, Monthly Report February 1970, p. 8.

Diagram 12 **Order Position of Industry**



1 Trend estimated on basis of linear regression on time over period 1959-1969.
Sources: IFO, OECD, Statistisches Bundesamt, DIW.

almost half a year's output, and investment orders remained at a high level in the first months of the year. Given the pressure on capacity and the sharp rise in labour costs, incentives to invest probably remain strong, and the financial constraints are not likely to be too severe in the current half-year, since the banks still have large unused credit commitments to the enterprise sector. Business fixed investment might therefore rise even faster than suggested by the IFO survey.

Other types of fixed investment may, however, rise less fast in the first half of the year. The increase in public capital expenditure will probably be modest, given the decision to postpone spending as far as possible to the second half-year. Residential construction is probably on a declining trend, and bad weather in the first quarter has exacerbated the fall.

Consumers' demand, which accelerated last year, is likely to remain strong, probably rising at an annual volume rate of 9-10 per cent in the first half of the present year. Since the early autumn of 1969, wage settlements have typically provided for increases of 11-12 per cent a year[1]. Employment is still rising. Moreover, given the change in the income distribution in favour of wage incomes, the personal savings ratio may decline somewhat. Even though the rise in consumer prices is accelerating, the increase in the volume of private consumption is likely to be substantial. The growth of public consumption, however, is likely to be more moderate; the general government budget points to a 3 per cent increase (annual rate) in the first half of 1970.

On this basis, final domestic demand (excluding stocks) is likely to remain buoyant in the current half-year, possibly rising at an annual rate of 7½ to 8 per cent. According to recent IFO surveys a considerable number of businessmen still consider their stocks to be too low (Diagram 11). A major increase in stockbuilding would seem unlikely, however, given the strong demand situation likely to prevail in the next few months and the tightening of credit conditions. Moreover, a sizeable increase in inventory accumulation would entail a rise in the stock/output ratio to a level which would be out of line with past experience.

Considerable uncertainty surrounds the prospect for exports. The growth of German export markets is likely to be more modest than earlier, given the outlook for world trade, and particularly for United States'

1 Taking into account the recent change in Social Security Legislation, compensation of employees may rise by some 16 per cent between 1969 and 1970. (From 1st January 1970 employers have taken over, from the Social Security system, payment of wages during sickness. The cost impact is estimated at some 2 per cent of the wage and salary bill).

Table 13 **Trends and Prospects**

Demand and Output

estimates and forecasts, volume seasonally adjusted, per cent (at annual rates)

	1968 Current prices (DM Bill)	1968	1969	1970	1969		1970		1971
					I	II	I	II	I
Private consumption	301.1	4.0	7.7	8½	8½	7½	9½	7¼	4½
Public consumption	83.6	–0.9	4.1	4	4¼	5¾	2¾	5	4½
Fixed investment	124.8	8.0	12.4	8	10¾	10½	8½	5	–1
Final domestic demand (excluding stocks)	509.5	4.2	8.3	7¾	8¼	8	8	6	3
* plus change in stock building	11.0	2.8	0.4	–½	¾	0	–½	–1¼	–¾
* plus change in foreign balance	18.0	0.2	–0.3	–1¼	–1¾	¾	–2¾	–¼	–¼
GNP at market prices	538.5	7.2	7.9	5½	7	8¼	4½	4½	2
GNP implicit price deflator	100.0	1.5	3.4	6½	2½	5½	7½	5	3½
Industrial production		13.4	13.0	8	16.1	7½	10	5	2

* The yearly and half-yearly rates of change refer to changes in stock building and in the foreign balance expressed as a percentage of GNP in the previous period. These two figures plus the rate of change of final domestic demand broadly equal the rate of change of GNP. In practice, however, divergencies are possible because the rate of change of final domestic demand is not expressed as a percentage of GNP.

imports. The inflow of foreign orders fell sharply after the DM revaluation. Nevertheless, new orders continued to exceed deliveries up to March (latest date available), when the volume of orders in hand may have corresponded to about 5-6 months' output by the export industries. This suggests that exports will continue to rise in the first half of the year, although at a much slower rate than earlier. Imports are likely to rise fast as aggregate demand is expected to increase much more than productive capacity and as the effects of revaluation work through. Real GNP may be growing at an annual rate of 4½ per cent in the first half of 1970; taking into account the loss of output because of bad weather in the early months of the year—amounting to perhaps ½ per cent of GNP—this would be roughly in line with the growth of capacity (estimated at nearly 5 per cent). Pressures on resources are therefore likely to remain severe in the current half-year, and the rise in prices may remain strong.

In the second half of 1970, however, the strains could begin to ease somewhat. The economy should be feeling the effects of the revaluation and of tighter monetary conditions on domestic demand. Given the marked change in the foreign balance forecast for the first half of the year, the further deterioration in the second half may not be very considerable; the easing of demand pressures should to some extent offset the revaluation impact. Business fixed investment demand could weaken, influenced by tighter credit conditions and the greater uncertainty in the demand outlook both at home and abroad. But no actual fall in the enterprise sector's capital expenditure would seem likely as the flow of investment orders—which at the present stage of the cycle may precede investment outlays by more than 6 months—has remained high in recent months. The increase in manufacturing investment indicated by the IFO survey for the year as a whole, together with the increase forecast for the first half of the year, suggests that the further real increase in business fixed investment may slow down markedly in the second half. Public investment and residential construction may show some rise since both, as noted, are likely to have been depressed in the current half-year. Nevertheless, total fixed investment is not expected to rise much. Private consumption is likely to be the most dynamic demand element; the increase may decline somewhat from the very high rate which seems to have prevailed in the first half of the year, reflecting little further growth in employment and, perhaps, some slowdown of the rise in wages. The growth of final domestic demand, although slowing down, may still amount to some 6 per cent (annual rate).

It is possible that inventory accumulation will be down in the second half of the year. The stock/output ratio should be back to a more normal level. Indeed, as the cash flow of the enterprise sector is likely to come under pressure and bank credit will be more difficult to obtain, some

pruning of inventory levels might take place in some sectors. The size of the possible decline in stockbuilding is obviously uncertain, but even with some further deterioration of the foreign balance, growth of real GNP might continue an annual rate of around 4½ per cent in the second half of the year, but with the underlying trend showing a marked deceleration[1].

Under these conditions, demand pressures could begin to ease somewhat in the second half of the year. The rise in employment is likely to level off. It is uncertain how quickly the easier demand situation will begin to affect prices and wages. Past experience both in Germany and other countries suggests that prices and wages react with some lag to changes in demand trends. While a more stable tendency in industrial producer prices would seem likely by the end of the year, the rise in consumer prices could remain significant, given the strong upward trend in produces prices in the current half-year. How far, in this setting, the present rapid increase of wages could be reduced would depend importantly on the degree of success enjoyed by the government's incomes policy.

Realisation of the present forecast may imply a marked increase in the borrowing requirements of the enterprise sector. On the assumptions adopted, including the estimated effects of the revaluation, the current external surplus may decline by DM 4 billion. Household savings may rise by DM 3½ billion and government financial savings by DM 5-6 billion (Table 14). The implication is that the financial deficit of the enterprise sector would rise by about DM 15 billion. Given the existing low level of the banks' secondary reserves and the possibility of a further reduction of these reserves due to " market forces " (including the budget), financing of a net financial deficit of this size in domestic money and capital markets could be difficult. The increase of the banks' minimum reserves in respect of incremental foreign liabilities to about 40 per cent—which could be raised to as much as 100 per cent—will probably serve to restrain the inflow of foreign funds to the banks[2]. But as the cost of domestic overdraft credits has increased sharply in recent months to about 11-12 per cent it may be attractive for non-bank enterprises to borrow at the substantially lower rates obtaining in the Euro-market.

1 Allowing for the effects of the hard winter on the construction industry, and special fiscal measures (see p. 34, the underlying rate of growth of GNP might be as follows:

	1970 I	1970 II
Percentage change at annual rate	5½	3¼

2 Minimum reserve requirements of this order imply that the cost to German banks of borrowing in the Euro-market has increased from, say, 9 per cent to 15 per cent.

Table 14 **Financial surplus (deficit) of the principal economic sectors**
(DM billion, current prices)[1]

	1968	1969	1970
Private households	35¼	40	43¾
Government	–4	9	14½
Enterprises	–21	–42¼	–55¾
Foreign sector	–10¼	–6¾	–2½

1 Secretariat estimates and forecasts.

Since policy measures affect demand only with a certain lag, some view on the demand situation likely to prevail in the first half of next year needs to be formed before demand management strategy can be decided upon. Experience in Germany and other countries has shown that there is considerable flexibility in the monetary system and that, at least in certain situations, the braking distance of restrictive monetary policy measures may be long. But given the degree to which bank liquidity and bond market activity have already been reduced, a marked slowdown of the credit expansion is likely. Thus, in view of the probability of a sharp increase in the financial deficit of the enterprise sector in the present year, it seems reasonable to assume that on the basis of unchanged monetary policy there would be some tendency by enterprises to reduce investment in inventories and fixed assets. The fact that business fixed investment has reached a very high level in relation to its long-term trend may increase the possibility of an adjustment, pointing to a levelling-off in the first half of 1971. Residential construction could be expected to fall, continuing the downward trend already apparent in 1969. On these assumptions total fixed investment could cease to grow and possibly turn down. Although world trade may be reviving, the rise in German exports could remain relatively weak because of the revaluation and the sharp increase in labour costs. Private consumption is likely to react with the normal lag and remain relatively buoyant, because although employment could fall, real income will be supported by a further increase in wages and more stable price trends. But the rise in real GNP, on the basis of existing policies, could be rather small. The rise in prices should become much more moderate.

Balance of Payments Prospects

In the present year, with domestic demand remaining strong through most of the period and the revaluation effects becoming felt, the current account surplus should fall sharply. The impact of the parity change is

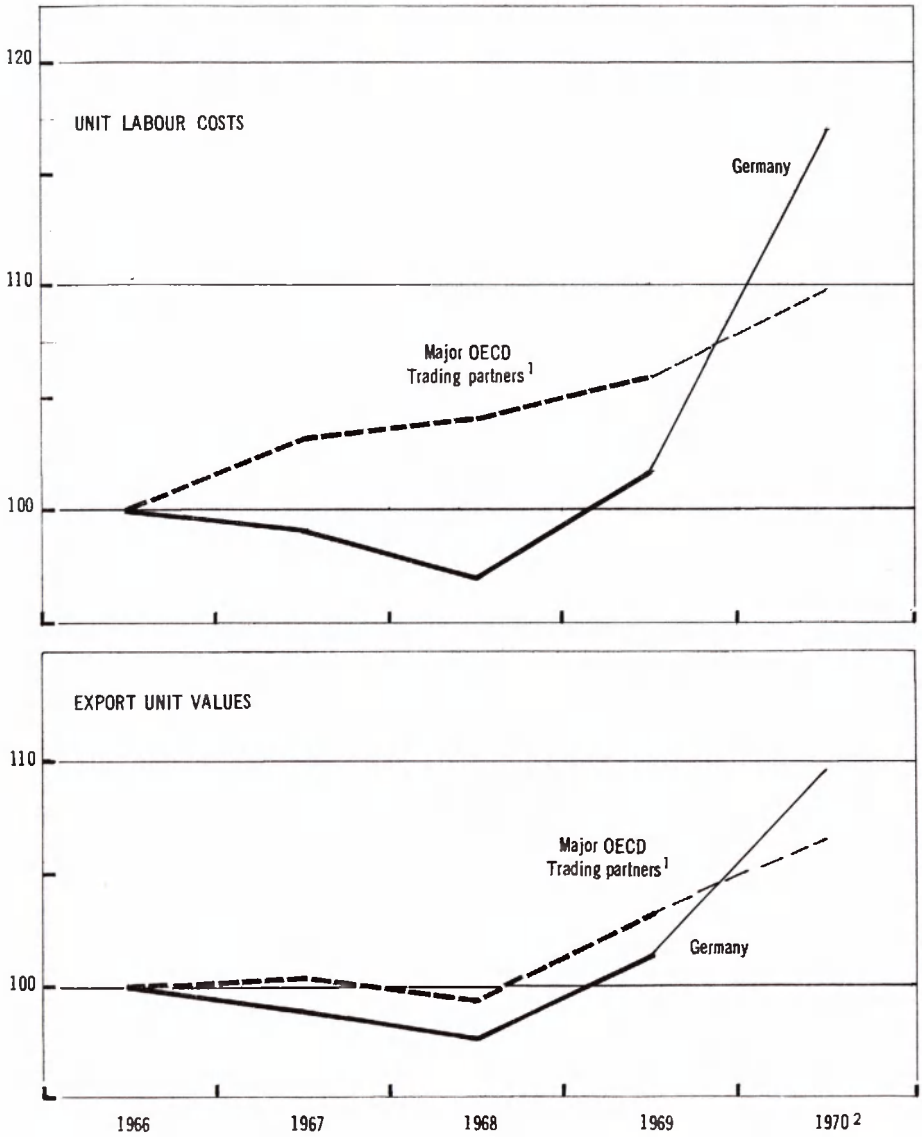
difficult to assess. Given the commodity composition of German exports, notably the importance of investment goods[1], for which the short-term price elasticity may be low, it is possible that, in the short run, the loss in volume will be less than the increase in foreign currency proceeds. Nevertheless, the existence of the border tax measures for some nine to ten months before the revaluation points to somewhat larger short-run effects than would have occurred if the parity change of October had been the only factor. By the end of 1970, then, and taking also into account the sharp rise in labour costs, it may be reasonable to expect some decline in German market shares. The substantial deterioration of the relative German competitive position resulting from the revaluation and the rapid rise in unit labour costs is illustrated in Diagram 13. Imports should rise sharply both because of the revaluation and the strong pressure on resources likely to prevail during most of the year. Allowing for a probable further increase in the deficit on services and transfers, the current account surplus may fall from $ 1¾ billion in 1969 to $ ½ billion in 1970[2]. For the balance of goods and services, this would imply a surplus equivalent to about 1½ per cent of GNP, roughly in line with the medium-term official balance of payments aim (1½-2 per cent).

The decline in the current account surplus is likely to be particularly sharp in the first half of the year. The rise in exports should slow down markedly from the level they had attained in the second half of last year when deliveries were particularly high in anticipation of the revaluation. The current account surplus could come down to a seasonally adjusted annual rate of $ 0.7 billion (from one of $ 2.1 billion in the second half of 1969). A further but more moderate decline can be expected in the second half of the year, if the rise in imports slows down in response to the easing of domestic demand pressures. By the beginning of 1971 the current account may not be far from an equilibrium position.

The capital account is likely to change markedly this year. The factors which produced large long-term capital outflows during 1968/69 have already changed importantly. The banks' liquidity position is now tighter than at any time during the 1964-66 squeeze. With yields on recent issues of foreign-DM-bonds issues now up to some 9 per cent and the interest rate differential towards the Euro-bond market considerably narrowed, the cost-incentive to borrow in Germany has virtually disappeared. Although the banks may still have some outstanding commitments to lend abroad, it is likely that the outflow on private account will slow

1 54 per cent of German exports in 1969.
2 Of the expected decline in the current account surplus, the revaluation has been estimated to be responsible for almost half, i.e. about $½-¾ billion.

Diagram 13 Relative Trends of Costs and Prices
(1961 = 100, dollar basis)



1 Weighted by productive capacity and importance for German trade.
2 Secretariat forecast.
Source: OECD.

down significantly. If the coupon tax is abolished, as proposed by the Government and the German enterprises begin to raise long-term funds abroad, it may be expected that the net outflow of long-term capital might fall considerably in the course of the year. Assuming that tight monetary conditions are being maintained, the short-term capital account (including errors and omissions) is likely to see sizeable inflows this year. After the recent increase in the discount and Lombard rates, short-term rates are somewhat higher than in the Euro-dollar market, and the financial system is likely to be exposed by heavy pressure in the months ahead. Overall, then, the capital account could show a net inflow and official reserves begin to rise again.

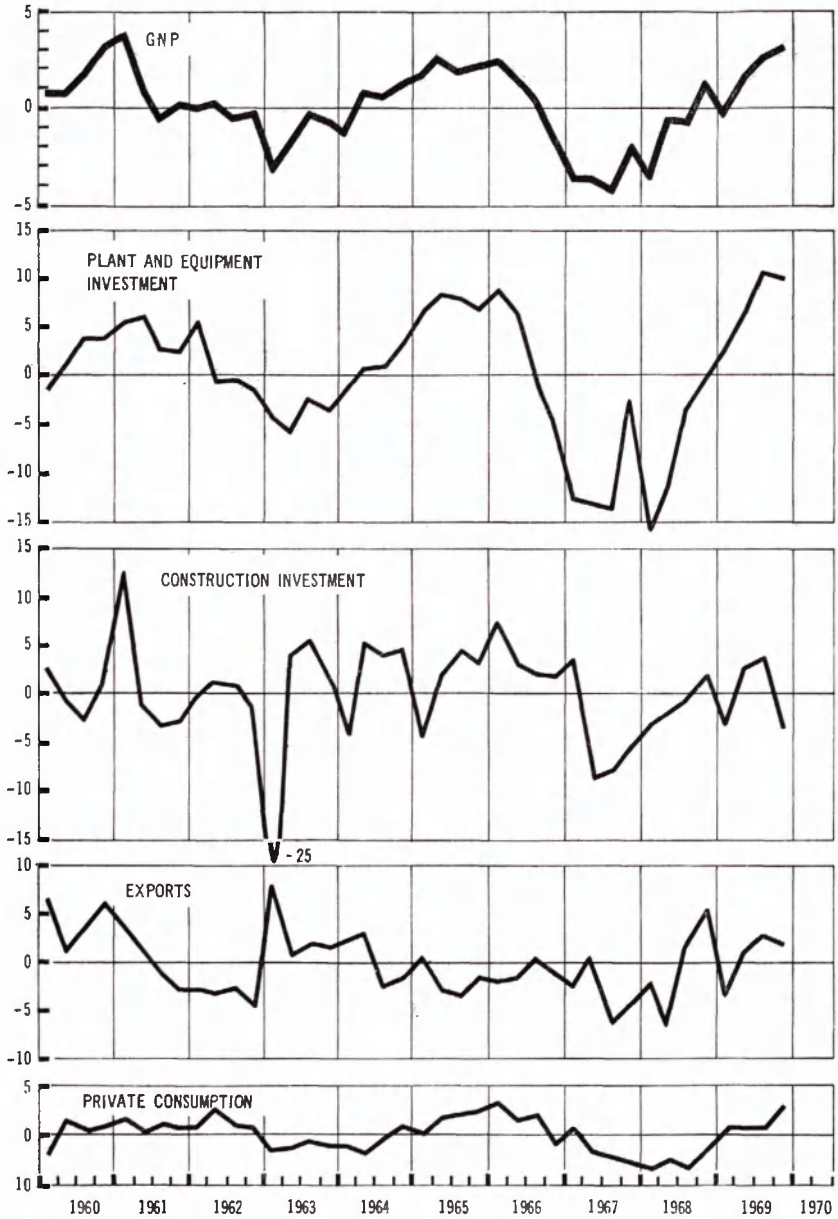
Conclusions

The extent to which output has been able to respond to the sharp increase of demand—real GNP has been rising at an annual rate of 7-8 per cent over the last two years—has largely reflected absorption of the slack created during the 1966-67 recession. Since the fourth quarter of 1965, when the previous business cycle reached its peak, real GNP has risen an annual rate of some 4¾ per cent, the same as during the previous cycle (first quarter 1961 to fourth quarter 1965). The rate of growth of labour productivity, however, accelerated from 4½ to 5 per cent; and in 1969 and 1970, with fixed investment well above the long-term trend, capacity is probably growing faster than its trend rate of the 1960's.

During the latter half of the 1960's economic policy has failed to achieve a steady advance in activity. Cyclical fluctuations were more marked than in the first half, partly because policy changes have tended to be made too late. Since the 1966-67 recession, policy-making arrangements have been considerably improved; short-and long-term forecasting procedures have been extended, and the arsenal of fiscal policy weapons has been enlarged. But the revaluation was too long delayed to prevent overheating of the economy; so long as the currency was under-valued and a large external surplus persisted, disinflationary fiscal and monetary measures proved largely self-defeating.

As stated by the Government, a main task of economic policy in the present year is to restore better price stability without creating another recession. A return to more normal rates of resource utilisation, entailing for some time an expansion of aggregate demand below the rate of growth of capacity, is clearly necessary. The measures already taken to restrain the growth of total demand should probably prove sufficient for this purpose in the immediate future, although the uncertainties in both directions are such as to warrant a very close watch being kept on the leading indi-

Diagram 14 Cyclical Movement of GNP and Some Main Components[1]



1 The graphs show deviations of quarterly seasonally adjusted volume figures from an OECD Estimate of Trend.

Sources: OECD, Deutsches Institut für Wirtschaftsforschung.

cators during the next few months. A difficult problem for the policy-makers may, however, be to time subsequent relaxations of demand management policy appropriately —and to achieve an appropriate policy mix—so as to ensure that restraint does not—as in 1966—exceed its target. Another major uncertainty is whether demand management, by itself, can prove sufficient to restore the medium-term degree of price stability at which the authorities aim (a rise in the GNP deflator of not more than 2½-2 per cent a year) and how far concerted action between government, labour and employers to restrain costs and prices can bridge any gap which remains.

The existing policy mix, with major reliance on monetary policy, is not ideal. Given the political difficulty of taking adequate fiscal action to restrain demand, the present mix was probably inevitable. But so great an accentuation of monetary restraint has its drawbacks. When corporate liquidity is high, the braking-distance of monetary restriction is long, and fiscal restraint would probably have worked more quickly. On the other side of the coin, heavy reliance on monetary policy bears particularly heavily—once it begins to bite—on the various forms of investment, the most volatile component of demand, and this increases the danger of an eventual adjustment of unwanted strength. The external impact of a policy mix heavily slanted towards monetary restraint can also be unfortunate: to the well-known fact that increasing resort by business to foreign sources of finance can partly frustrate the domestic aims of policy must be added the danger that, for balance-of-payments reasons, such action can make a movement away from high interest rates by other countries more difficult.

Since a quick shift from monetary to fiscal restraint is probably not possible, the major immediate recommendation to the German authorities must be to avoid any overall relaxation of demand management policy. It is still an open question whether demand pressures will weaken sufficiently in the spring and summer to prevent an inflationary climate when the important wage negotiations begin again in the autumn.

But before the end of the year it may perhaps be necessary to relax monetary policy to prevent a serious weakening in the investment sector during 1971—or, alternatively, to prevent an embarrassingly large net capital inflow. Here, a difficult problem of judgment may be involved, again entailing the mixture of monetary and fiscal policy. It might prove desirable—should the wage round seem likely to result in inappropriately large increases—to accompany monetary relaxation by some temporary tightening of fiscal policy.

The apparent paradox implied by these recommendations—to relax monetary policy but to accompany (or, indeed, to precede) this by contrast-

ing fiscal action—stems from the time-lags before changes in monetary policy exert their full effect on demand. But the dangers of continued demand pressures in the next few months and of a serious weakening next year are not incompatible. As the forecasts outlined on page 43 suggest, unchanged policies, through their effect on fixed investment and stock-building, could bring the expansion to a virtual halt in the course of 1971. If this were to happen, the danger of a recession would increase. An economy brought to a standstill tends to become unstable. Unless forward momentum is re-established, downward forces may become dominant.

Decisions on the timing and size of policy changes may be particularly difficult in the next few months. It is important that allowance should be made for timelags between the adoption of measures and their demand effects, and that measures should be taken on the basis of leading rather than lagging indicators. Since prices are slow to react to a change in demand, to delay action until they were performing satisfactorily could seriously enhance the risk of recession. But Germany is well-equipped with relatively reliable forward-looking indicators, such as business tests in industry and order data for both industrial goods and construction projects, and reliance on these should reduce the risk of major mistakes.

Even if demand pressures should ease roughly as now envisaged, there is no certainty that a satisfactory price performance will be restored. In the past Germany has been relatively successful in combining full employment and a high degree of price stability. But the possibility cannot be excluded that wage claims will remain excessive from the point of view of price stability for some time to come. The German authorities have already operated an incomes policy, which enjoyed some success in the initial phase of the upswing. Indeed, one reason why the rise in wages in 1968 and early 1969 lagged considerably behind the general growth of incomes, was that wage decisions were based on low forecasts of the growth of national income. This, together with the overheating subsequently allowed to develop, probably rendered the recent sharp increase of wages inevitable. What is important now is to prevent the further adjustment of wages from going too far. The German authorities will no doubt wish to use the existing arrangements for operating an incomes policy (" concerted action ") in order to restore more stable conditions. Early success in this respect could shorten the period during which some margin of unused resources will have to be maintained and enhance the prospect for steady growth in the years to come.

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Annex I

CHRONOLOGY OF MEASURES AFFECTING MONETARY DEVELOPMENTS AND INTERNATIONAL CAPITAL FLOWS

November 1968

Minimum reserve ratio of 100 per cent imposed against increases after 15th November in banks' foreign deposit liabilities.

Authorisation required for banks to incur a range of foreign liabilities, except in settlement of " normal " operations. Ban on interest payments to foreigners extended to savings accounts.

Bundesbank's foreign currency swap operations temporarily extended to non-banks (to 2nd December only).

February 1969

Authorities cease open market operations in long-term government bonds.

Banks no longer required to obtain authorisation to accept foreign funds (see November).

Base date for computation of 100 per cent minimum reserve requirements against increase in foreign deposits (see November) now 15th November or 15th January.

March 1969

In order to relieve strain on the bond market Federal authorities announce that they would make no new bond issues until further notice.

Banks agree to establish informal " waiting list " for foreign issues on German capital market.

With economy threatened by overheating:

— Lombard rate (on short-term lending to commercial banks against collateral) increased from 3.5 to 4 per cent.

— Banks rediscount quotas reduced by about DM 2.6 billion effective 1st July.

April 1969

In view of growing cyclical tension, discount rate raised from 3 to 4 per cent and Lombard rate from 4 to 5 per cent.

To avoid upward pressure on domestic interest rates the (unofficial) Central Capital Market Committee decides to spread over a longer period foreign DM loan issues in the process of introduction by the banks.

Base date for computation of 100 per cent minimum reserve requirement against increases in foreign deposits (see November and February) now April 15th or 30th (Previous intention was to repeal measure in May).

May 1969

Effective 1st June, increase in banks' minimum reserve ratio against domestic liabilities by 15 per cent, and by 50 per cent against foreign liabilities up to level on 15th or 30th April (see April). Resultant increase in required reserves estimated at about DM 2.5 billion.

Foreign currency swap operations with the banks temporarily suspended (7th to 21st).

Second decision not to revalue Deutschemark.

June 1969

Discount rate raised from 4 to 5 per cent, and Lombard rate from 5 to 6 per cent.

July 1969

Law passed by Parliament permitting Bundesbank to require 100 per cent minimum reserves against banks' total foreign liabilities.

August 1969

Minimum reserve ratios raised by 10 per cent against domestic liabilities and foreign liabilities up to level of 15th or 30 April (see April). Measures absorb about DM 1.6 billion of bank liquidity.

Change in minimum reserve regulations designed to reduce the attractiveness of " currency-round-about operations " which do not produce additional short-term capital exports (round-about operations took place when the Bundesbank provided forward cover at rates below market rates).

September 1969

Special supplements to Lombard rate obligatory, when in the course of a month the Lombard credits extended to a bank exceed by a given factor its capital and published reserves.

Discount rate raised from 5 to 6 per cent (highest rate since the Korean boom). Lombard rate raised from 6 to 7 per cent and special supplements abolished.

Bundesbank suspends swap operations with German banks.

Bundesbank suspends operations to maintain Deutschemark exchange rate within 1 per cent of parity with the Dollar.

October 1969

Deutschemark revalued by 9.3 per cent (27th October 1969).

November 1969

Abolition of the 100 per cent incremental reserve requirements on foreign deposits, complete unification of reserve requirements on foreign and domestic deposits (from now on reserve requirements apply to all deposits together), and reduction of 10 per cent in minimum reserve requirements. These measures released about DM 2 billion of bank liquidity and were designed as a partial offset to the sharp fall in bank liquidity resulting from reversal of speculative money flows.

To relieve pressure on the bond market, the Central Capital Marke. Committee agreed to limit monthly total of foreign issues to DM 0.3 billiont

December 1969

Central Bank decides on a temporary 10 per cent cut in minimum reserve requirements in order to relieve year end pressure on the money markets. Measure releases DM 1.8 billion of bank liquidity.

Increase of the Lombard rate to 9 per cent. Measure aims at protecting the domestic money market from too sharp an outflow resulting from the money exports of the banks.

Ban on interest payments by banks to foreigners suspended.

To relieve pressure on the bond market, the Central Capital Market committee agrees on issue pause for " once-only issuers ".

January 1970

Temporary cut in minimum reserves suspended (see December).

February 1970

Issue pause suspended (see December 1969).

March 1970

Increase of the discount rate from 6 to 7½ per cent. Lombard rate raised from 9 to 9½ per cent. Minimum reserve requirements on increases in foreign liabilities above either February average or March 6th level raised by 30 percentage points above the corresponding ratio for domestic liabilities (effective April 1st).

Annex II

GERMAN AVERAGE VALUE AND PRICE DATA IN FOREIGN TRADE

The very large price increases as reflected in the price indices recorded by German exports during 1969 ($16\frac{1}{2}$ per cent in dollars between December 1968 and December 1969) raises the question of the extent to which these will be reflected in increases in average values in 1970. For forecasting the movement of average values, it would be useful, then, if one could establish a relationship between the price and average value series in German foreign trade, especially if a regular lag relationship could be shown.

Diagram A shows the development of the two indices for both imports and exports since 1962. In the case of exports the price index has been consistently above the average value one since the end of 1963, and the gap between the two has grown almost constantly. (A semi-log regression between the ratio of the two indices and time gives a 0.9 R^2—the elasticity between prices and average values implies a 1 per cent increase in the ratio per year). Explanations for this different behaviour of the two series can be sought in any of the following three reasons :

(*a*) Differences in the commodity coverage of the two indices;

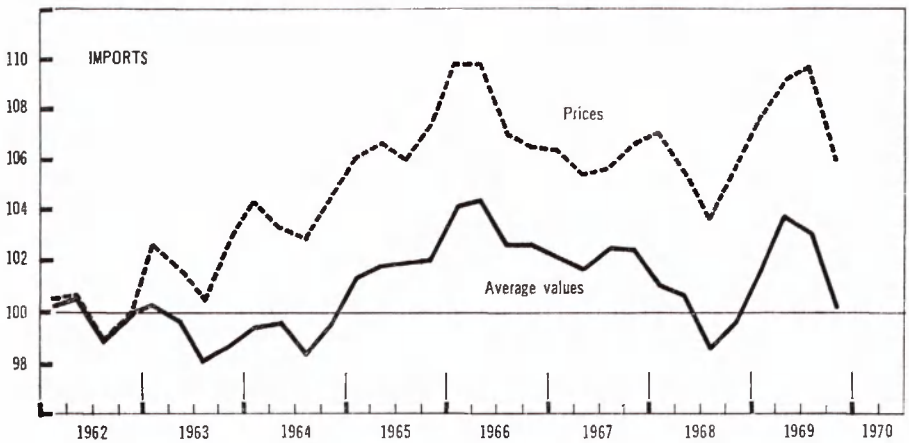
(*b*) Differences in the time periods covered by the two indices;

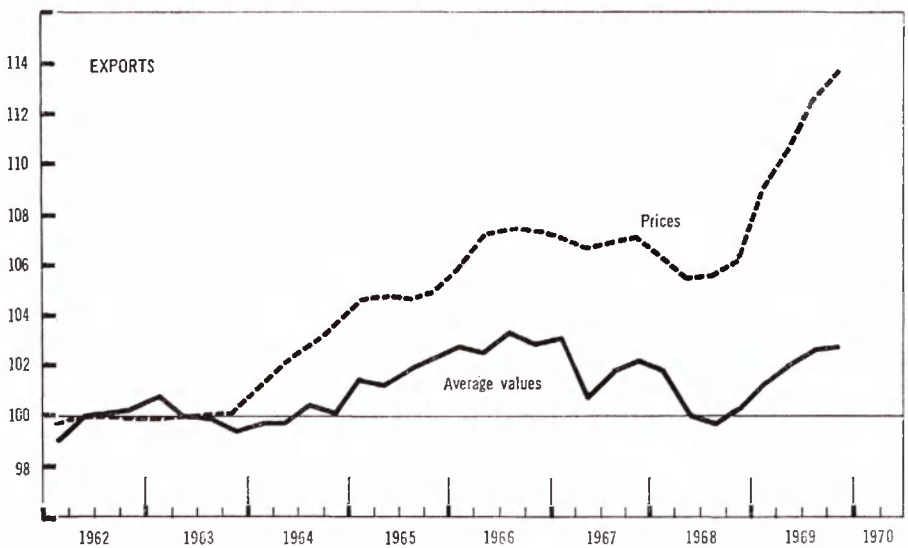
(*c*) Changes in the commodity pattern of German exports implying changes in weights in one index not reflected in the other (the price index being a fixed weight Laspeyres index, the average value index a moving weights Paasche one).

In terms of coverage in the 1962 base year, both indices seem to be very similar. The price index uses 3,600 price series covering 97 per cent of all exports[1], the average value index, which covers " practically all exported goods "[2], uses some 6,000 six-digit commodity classification groups[3]. Thus coverage seems nearly complete and sufficiently detailed in both cases.

1 *Wirtschaft und Statistik*, May 1967, p. 286.
2 *Wirtschaft und Statistik*, October 1966, p. 620.
3 Ibid, p. 621.

Diagram A **Indices of Price and Average Value**
(1962 = 200)





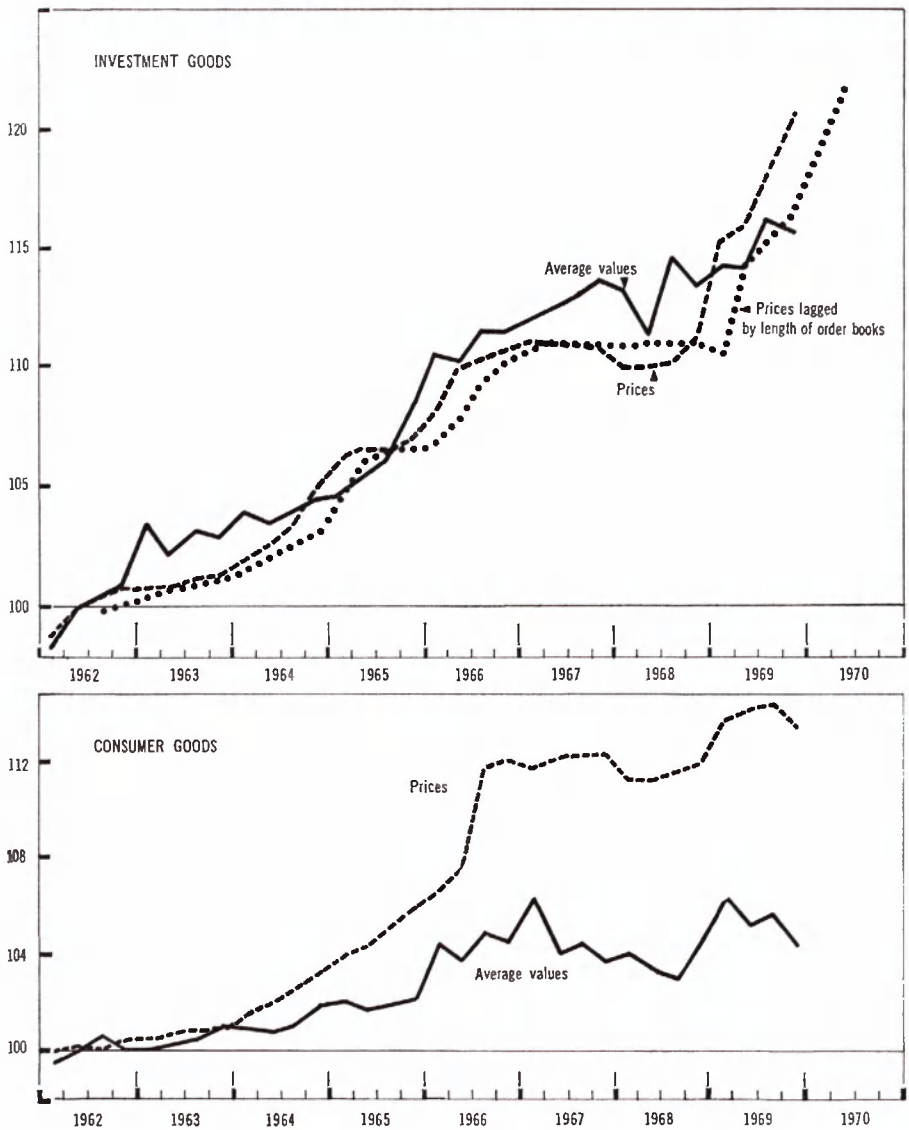
Source: Statistisches Bundesamt.

Conceptually the two indices differ in the time periods they cover—the price index registers price quotations when contracts are entered into; the average value index is based on values and volumes that actually cross the border. In theory, therefore, price movements should precede similar movements in average values, apart from any cancellation of orders. In practice, however, it is difficult to define a lag relationship. A linear regression between the two indices gives low correlation coefficients whether regular lags are introduced or not. A one-quarter lag regression between prices and average values fares marginally better than one without lags but the difference is extremely small. For investment goods an attempt was made to improve the correlation by the use of lags varying with average lengths of order books (see Diagram B) but even this device does not transform the price series into a usable leading indicator of the average value series. This may reflect the fact that the lag structure is complex, i.e. when the *average* order book length stretches to 6 months, this this may imply for some sectors order books of twelve months or more. The effects of an increase in the price index could, then, show up in the unit value index over a very long period. Even if definite lagged reactions could be established, however, use of the price index as a leading indicator would have to take account of the widening gap between the two indices.

This gap results from a " change in commodity composition " (including the introduction of new products). In order to tackle the problem in greater detail, the available indices for two main categories of German exports (consumer and investment goods) have been plotted in Diagram B[1]. The investment goods indices are erratic and heavily influenced by cyclical developments i.e. both in 1964-1965 and in 1968-1969 prices rose faster than unit values as delivery periods lengthened. It is only in 1969 that there are close similarities between the sector's and the global indices. For consumer goods, on the other hand, a large and growing gap has developed since end-1963. Some definite changes in weights can be observed in the time span under consideration even within the rather crude breakdown which is available (Table A). This is especially noticeable for two main categories: textiles and clothing on the one hand and passenger cars on the other—the former's share in consumer good exports went up by 7 per cent, between 1962 and 1969, the latter fell by just over 10 per cent. At the same time textiles average values declined by 2½ per cent while car average values rose by 6 per cent. Though such a shift can hardly explain the whole of the 6½ per cent discrepancy existing between the two indices in 1969 it shows that compositional changes have played some role. And similar changes

1 The two categories are looked at from the end-use and not the production side; they thus cover rather less of total German exports than could be expected (some 42 per cent in both 1962 and 1969).

Diagram B Indices of Export Prices and Average Values
(1962 = 100)



Source: Statistisches Bundesamt.

have probably taken place in the semi-manufactures category for which average value indices are not available. As for any remaining unexplainable discrepancy, it can probably be imputed to the inaccuracies of statistical observations.

Table A **Consumer goods exports**
Percentages

	Share in Total	
	1962	1969
Total exports of consumer goods	100.0	100.0
Textiles, clothing, footwear	9.0	15.8
Passenger cars	36.0	25.7
Electrical consumer goods	14.1	15.9
Furniture	1.5	4.6
Pharmaceutical products	6.8	9.5
Other	32.6	28.5

Since detailed sub-indices and complex lag relationships would apparently be needed to forecast unit values from prices, it would seem that, in practice, the price index can be used only as an indicator of the broad direction in which the average value index is likely to move and does not lend itself easily to quantification. The recent sharp increases in the price index suggest continued upward movement of the average value indices for some time ahead but the timing of these increases cannot be derived with any certainty. For investment goods, at least, past conjunctural experience might suggest that the German economy has now reached a stage where average values of exports may begin to increase more rapidly.

For imports the general conclusion on the medium-term development of the indices is not very different. The price index has been consistently higher than the average value one since 1963, but movements have been parallel and there is even less evidence of any lags. The ratio between the two indices has been fairly constant, though a scatter diagram of this ratio with time shows some evidence of a cyclical pattern, prices being more responsive to both upward and downward movements in import demand.

The reasons for a once-and-for-all upward shift in the price index in 1963 are not easy to determine. Compositional changes were quite marked between 1962 and 1963. Wheat imports fell by 50 per cent and the share of food imports in total imports was reduced by some 4 per cent (while average values went up); on the other hand imports of e.g. oil and chemicals

rose sharply while their average values fell. But similar, if not quite so rapid, shifts have been at work since 1963, and there is no evidence of any further change in the relationship between the two indices. It is possible that a more detailed commodity by commodity analysis could throw some light on this phenomenon. In any case, the very nearly parallel movement in the two indices implies at least that changes in the price index can provide some guide for changes in the average value one for the same period of time.

STATISTICAL ANNEX

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Table A **Gross National Product**
DM billion

	Current prices				1962 prices			
	1966	1967	1968	1969	1966	1967	1968	1969
Expenditure:								
Private consumption	277.9	284.2	301.1	332.3	247.4	248.9	258.9	278.8
Public consumption	76.5	81.0	83.6	93.2	60.8	62.8	62.3	64.8
Gross fixed asset formation	126.3	114.4	124.8	146.9	113.4	103.8	112.1	126.5
Machinery and equipment	52.6	48.0	52.9	68.0	49.0	44.6	49.3	61.6
Construction	73.7	66.4	71.9	78.9	64.4	59.2	62.8	64.9
Change in stocks	3.6	–1.3	11.0	13.5	3.4	–1.2	11.0	13.3
Exports of goods and services	102.3	110.5	124.4	141.6	97.4	105.6	119.8	134.1
Imports of goods and services	95.8	94.3	106.4	126.4	90.6	89.4	102.6	119.2
Gross national product at market prices	490.7	494.6	538.5	601.0	431.7	430.5	461.5	498.2
Origin:								
Agriculture, forestry, fishing	20.7	20.2	20.5	21.8	18.3	20.2	21.1	21.6
Industry, building, energy, water	262.1	257.2	288.3	327.4	242.2	236.4	259.0	285.8
Trade, transport, communications	94.7	96.6	102.5	111.6	82.5	82.6	86.8	93.6
Other services	114.6	122.1	131.7	140.6	90.0	92.7	95.4	97.7
Gross domestic product at market prices	492.1	496.1	539.2	601.4	433.0	431.9	462.2	498.7

Sources : Statistisches Bundesamt and OECD Statistical Questionnaire.

NOTE Figures may not add up due to rounding and, in 1968, the changeover from Turnover Tax to TVA.

Table B Distribution of National Income
DM billion, current prices

	1965	1966	1967	1968	1969
Compensation of employees	230.0	247.6	248.0	266.0	299.4
Employers' contributions to social security	23.8	26.4	27.0	29.8	34.4
Employers' social security payments and wage tax	34.6	39.5	40.2	45.5	54.4
Net wages and salaries	171.6	181.7	180.8	190.8	210.6
Distributed profits, etc.	72.9	76.9	78.0	87.2	99.5
Current transfers from the state	58.3	64.0	70.4	73.6	79.0
Less: Interest, on consumer debt, etc.	8.3	9.6	9.6	9.7	11.3
Disposable income[1]	294.4	312.9	319.6	341.9	377.8
Income from property and entrepreneurship	125.3	129.5	127.1	149.3	160.0
Direct taxes, etc.	29.2	30.2	29.9	32.5	36.1
Net income from property and entrepreneurship	96.1	99.3	97.2	116.8	123.9
of which:					
Distributed to households	71.4	75.2	76.4	85.5	97.5
Distributed to state	3.8	3.7	2.9	3.3	3.4
Retained	20.8	20.4	17.9	28.0	23.0
National Income	355.3	377.1	375.1	415.3	459.4

1 Exluding retained profits.
Source : Statistisches Bundesamt.

Table C **Employment and Labour Market**
Thousands

	1966	1967	1968	1969	1968				1969			
					I	II	III	IV	I	II	III	IV
Labour force	27 243	26 751	26 665	27 001	26 585	26 511	26 668	26 904	26 781	26 902	27 116	27 203
Employment, total	27 082	26 292	26 342	26 822	26 000	26 198	26 480	26 690	26 448	26 750	27 010	27 079
German nationals	25 838	25 278	25 323	25 456	25 049	25 183	25 390	25 670	25 268	25 430	25 550	25 539
Foreign nationals[1]	1 244	1 014	1 019	1 366	951	1 015	1 090	1 020	1 180	1 320	1 460	1 540
Self-employed, and family workers	5 212	5 112	5 012	4 902	5 026	5 018	5 006	4 998	4 962	4 922	4 882	4 842
Wage and salary earners	21 870	21 180	21 330	21 920	20 974	21 180	21 474	21 692	21 530	21 844	22113	22192
Wage and salary earners in manufacturing and building	9 568	8 932	9 054	9 495	8 797	8 971	9 161	9 266	9 250	9 443	9 607	9 680
of which: Basic materials	1 771	1 670	1 682	1 750	1 643	1 673	1 701	1 710	1 715	1 739	1 767	1 779
Investment goods	3 617	3 385	3 466	3 741	3 380	3 414	3 495	3 577	3 638	3 693	3 773	3 859
Consumer goods	2 040	1 902	1 908	1 990	1 878	1 886	1 916	1 952	1 968	1 979	1 997	2 015
Food, drink, tobacco	521	507	505	510	496	497	513	516	503	502	517	518
Building construction	1 619	1 467	1 427	1 505	1 400	1 501	1 536	1 510	1 426	1 530	1 554	1 508
Hours worked in manufacturing (*million*)	11 565	10 441	10 826	11 481	2 641	2 638	2 686	3 016	2 809	2 820	2 836	3 016
Hours worked in building (*million*)	2 793	2 475	2 472	2 440	456	677	702	638	419	692	723	605
Short-time workers	16	143	10	1	30	10	3	1	3	1	1	1
Unemployment, seasonally adjusted	161	460	324	179	359	341	293	229	205	161	162	146
Vacancies, seasonally adjusted	540	302	488	747	386	465	532	613	679	755	753	837

1 Quarterly figures relate to end of quarter, annual figures are averages of end of quarters.
Sources : Statistisches Bundesamt, Main Economic Indicators, OECD.

Table D **Industrial Production, New Orders and Deliveries**
Seasonally adjusted. Indices

	Base year	1966	1967	1968	1969	1968				1969			
						I	II	III	IV	I	II	III	IV
Production:													
Industry, excl. building	1963	116.3	113.8	127.5	144.0	119.8	126.6	129.0	135.4	139.6	143.7	143.3	149.9
Mining and quarrying	—	94.3	88.6	91.9	95.1	91.4	91.8	90.8	94.1	94.9	95.7	92.7	97.5
Capital goods	—	114.4	106.4	119.7	142.1	110.2	118.8	121.9	129.8	135.9	142.9	141.5	149.4
Consumer goods[1]	—	116.2	111.1	127.1	141.5	117.8	128.6	127.8	135.5	137.5	143.3	139.8	146.1
Materials	—	123.9	127.2	145.4	162.7	137.2	143.3	147.6	154.0	157.0	160.7	163.5	170.4
Building, not seasonally adjusted	—	119.4	111.0	117.6	120.7	81.4	137.6	128.5	125.1	78.8	145.9	137.5	122.8
New orders in manufacturing[1]:	1962												
Total manufacturing, all markets	—	128	127	147	180	133	143	152	159	170	178	189	184
domestic	—	121	117	134	164	122	131	139	145	153	161	171	172
foreign	—	157	174	204	250	184	196	212	224	246	252	266	236
Capital goods, all markets	—	131	130	155	204	139	150	163	167	192	205	222	198
domestic	—	122	116	137	183	122	131	143	150	167	181	196	186
foreign	—	157	172	207	261	187	197	215	228	254	261	285	245
Consumer goods, all markets	—	124	123	139	155	127	136	140	151	152	146	154	167
domestic	—	121	119	133	147	122	131	134	143	144	138	147	157
foreign	—	160	176	209	261	185	200	219	233	253	267	269	254
Deliveries by manufacturing[1]:													
Total manufacturing, all markets	—	128	124	134	158	123	130	136	146	147	154	162	170
domestic	—	123	116	122	146	112	119	124	134	134	138	148	162
foreign	—	150	164	186	215	170	177	192	205	200	220	223	217
Capital goods, all markets	—	131	124	133	164	121	130	139	143	153	162	173	166
domestic	—	124	114	117	146	105	115	120	127	136	140	153	156
foreign	—	148	155	181	210	164	175	190	194	187	225	222	206
Consumer goods[1], all markets	—	128	122	132	149	122	129	131	144	139	144	147	166
domestic	—	125	118	125	141	117	123	125	136	132	135	139	159
foreign	—	161	175	209	254	187	199	210	239	236	260	260	261

1 Excluding food, drink and tobacco.

Sources : Main Economic Indicators, OECD, IFO Schnelldienst.

Table E Wages and Prices

Indices

	Base year	1966	1967	1968	1969	1968 I	1968 II	1968 III	1968 IV	1969 I	1969 II	1969 III	1969 IV
Earnings and productivity in manufacturing:													
Hourly wage rates	1962	130.9	137.8	143.8	153.1	139.4	143.7	145.9	146.3	148.6	150,6	152.8	160.5
Hourly gross earnings	—	136.7	141.8	147.9	161.3	143.3	145.8	150.6	151.7	154.5	158.3	162.0	170.2
Output per hour worked	—	121.1	128.3	144.3	152.8	131.6	148.6	148.9	145.0	144.6	160.7	154.6	150.9
Labour cost per unit of output	—	114.6	113.7	110.5	113.5	108.4	111.8	111.9	109.8	108.4	112.3	115.2	117.4
Agricultural producer prices	1961-63	113.1	105.1	99.8	106.5	98.7	97.3	99.5	103.6	106.4	105.6	106.2	107.8
Vegetable products	—	106.0	94.3	85.2	97.6	85.8	87.8	82.1	86.3	96.5	103.3	94.7	95.7
Animal products	—	115.4	108.6	104.7	109.5	103.0	100.5	105.3	109.4	109.7	106.4	110.0	111.9
Industrial producer prices, home market sales[1]	1962	105.8	104.9	99.3	101.5	105.5	104.7	104.6	104.7	100.1	100.6	101.6	103.8
of which: Capital goods	—	108.6	108.3	102.4	106.9	101.9	102.1	102.7	103.0	104.5	105.3	106.7	110.9
Consumer goods[2]	—	106.7	107.3	101.2	102.4	101.4	100.8	101.0	101.5	101.7	101.9	102.4	103.6
Cost of dwelling construction[3]	—	118.5	115.9	120.8	127.2	119.9	119.9	121.5	121.9	122.6	125.7	128.0	132.6
Export prices, industrial goods	—	106.8	106.9	106.0	111.8	105.9	105.7	105.8	106.5	109.6	110.9	112.7	113.9
of which: Capital goods	—	109.7	110.9	110.4	117.2	110.1	110.0	110.2	111.2	115.1	115.7	117.6	120.1
Consumer goods[2]	—	108.4	110.0	109.4	111.8	109.1	109.1	109.4	109.8	111.5	111.9	112.2	111.4
Import prices, industrial goods	—	107.0	105.4	105.9	108.0	107.3	105.2	105.1	106.0	107.4	108.4	109.7	106.5
of which: Capital goods	—	109.0	107.7	108.1	108.6	108.6	108.0	107.9	107.9	108.8	109.6	110.0	106.1
Consumer goods[2]	—	99.0	101.9	102.6	100.5	104.7	101.0	101.3	103.3	102.0	100.7	101.2	97.9
The cost of living	—	122.8	114.4	116.1	119.3	115.7	115.9	115.9	116.9	118.3	119.3	119.5	120.2
Food, drink, tobacco	—	112.0	112.0	111.1	114.2	111.4	111.4	110.2	111.2	113.2	114.9	114.5	114.1
Non-food	—	113.4	116.3	120.0	123.5	119.0	119.4	120.4	121.2	122.5	122.8	123.6	124.9
of which: Rent	—	129.7	138.2	148.4	161.9	143.9	147.3	149.9	152.6	158.9	161.6	163.0	164.0

1 Excluding turnover tax from 1968.
2 Excluding food, drink and tobacco.
3 Data relate to the end of February, May, August and November.

Source : Statistisches Bundesamt : Sachverständigenrat zur Begutachtung der gesamtwirtschaftlichen Entwicklung.

Table F **Volume of Money, Monetary Capital Formation and Bank Liquidity**
DM billion, end of period

	1965	1966	1967	1968 I	1968 II	1968 III	1968 IV	1969 I	1969 II	1969 III	1969 IV
Volume of money	83 680	84 195	95 721	86 459	91 161	93 207	103 050[1]	94 106	100 056	102 750	114 251
Note and coin circulation[2]	29 652	30 876	31 507	31 139	32 064	31 920	32 587[1]	32 580	33 611	33 866	34 689
Sight deposits of German non-banks	48 873	48 743	56 414	49 989	53 423	55 043	60 879[1]	54 971	59 148	60 415	64 740
Time deposits under 3 months	5 155	4 576	7 800	5 331	5 674	6 244	9 584[1]	6 555	7 297	8 469	14 822
Banks' lending to domestic non-banks	319 602	347 805	380 536	386 826	397 584	407 405	420 606[1]	429 943	438 967	450 009	474 062
Bundesbank	12 805	13 097	13 765	11 458	11 287	10 922	11 691[1]	10 561	9 934	9 916	12 425
Credit institutions	306 797	334 708	366 771	375 368	386 297	396 483	408 915[1]	419 382	429 033	440 093	461 637
Public authorities	47 530	53 964	67 892	73 007	74 908	77 890	74 330[1]	75 650	75 123	76 428	75 709
Other non-banks	259 267	280 744	298 879	302 361	311 389	318 593	334 585[1]	343 732	353 910	363 665	385 928
Short-term	69 295	75 371	81 960	82 937	85 989	87 396	89 843[1]	92 258	94 782	95 694	104 181
Medium-and long-term	217 432	238 723	260 937	267 081	274 212	282 013	291 232[1]	298 270	305 617	315 160	328 127
Security holdings	11 514	11 873	15 024	16 432	17 222	18 302	19 168[1]	20 134	20 023	20 791	21 011
Other	8 556	8 741	8 850	8 918	8 874	8 772	8 672[1]	8 720	8 611	8 448	8 318
Domestic monetary capital formation	263 265	294 370	320 826	334 078	340 893	348 325	361 157[1]	375 299	383 944	392 780	398 705
Time deposits exceeding three months	91 237	103 241	108 454	113 274	115 117	117 621	124 200[1]	130 210	133 389	136 684	133 071
Public authorities	71 097	78 327	80 522	80 877	81 562	82 971	82 280[1]	83 359	84 432	85 454	86 419
Other non-banks	20 140	24 914	27 932	32 397	33 555	34 650	41 920[1]	46 851	48 957	51 230	46 652
Savings deposits[3]	109 758	126 029	143 677	149 802	152 762	156 520	166 358[1]	172 258	175 920	179 410	187 948
Bank bonds held by non-banks	42 479	46 188	47 875	49 297	50 124	50 986	48 272[1]	49 708	50 394	52 159	53 173
Capital and reserves	19 791	18 912	20 820	21 705	22 890	23 198	22 327[1]	23 123	24 241	24 527	24 513
Changes in bank liquidity (DM billion):											
Domestic liquid assets[4 5]				1 099	364	196	1 380	–2 799	–1 582	22	–3 406
Foreign liquid assets[6]				266	29	608	–166	1 871	2 885	–1 930	504
Unused re-discount quotas				–71	312	–228	1 436	–4 421	2 942	–2 384	–3 428
Total liquid holdings				1 294	705	576	2 650	–5 349	4 245	–5 192	–6 330
Less: recourse to Bundesbank Lombard credit				–191	–34	121	128	483	–151	–637	2 219
Change in total bank liquidity				1 485	739	455	2 522	–5 832	4 396	–4 555	–8 549
Freely usable reserves of credit institutions (DM billion)				31 666	31 662	33 308	37 193	28 627	34 305	33 033	19 790
(Per cent of total deposits)				14.0	13.5	13.6	14.0	11.3	12.9	11.9	6.8

1 Break in series.
2 Excluding bank holdings.
3 Including savings certificates.
4 Domestic treasury bills and non-interest bearing bonds, storage agency bills and prime bankers' acceptances.
5 Including excess reserves with the Bundesbank.
6 Short-term balances with foreign banks and foreign money market paper.

Source : Deutsche Bundesbank.

Table G **Merchandise Trade by Area**
DM million, monthly averages

	Total	OECD countries					Third Countries					
		Total	EEC	EFTA	North America	Other	Europe[1]	Africa	Central and South America	Asia	Australia and Oceania	Eastern Bloc
						Imports cif						
1965	5 871	4 297	2 221	1 011	842	223	99	428	396	343	57	243
1966	6 056	4 381	2 315	996	839	231	104	481	391	371	57	263
1967	5 849	4 223	2 303	916	792	212	90	481	370	376	51	251
1968	6 765	4 925	2 788	1 055	831	251	111	574	372	431	60	284
I	6 331	4 538	2 519	1 017	758	244	92	576	370	433	50	265
II	6 472	4 687	2 611	1 013	852	211	106	559	384	428	52	249
III	6 840	5 013	2 875	1 046	870	222	116	560	359	422	77	285
IV	7 417	5 464	3 146	1 146	844	328	132	600	576	439	60	338
1969	8 164	6 070	3 536	1 268	958	308	158	643	433	438	70	331
I	7 613	5 574	3 298	1 195	788	293	152	660	407	454	74	283
II	8 194	6 171	3 622	1 240	1 025	284	146	664	432	419	68	294
III	8 187	6 068	3 506	1 238	1 017	307	170	605	472	440	78	345
IV	8 663	6 467	3 718	1 401	1 000	348	162	643	422	439	61	401
						Exports fob						
1965	5 971	4 572	2 102	1 612	543	315	145	271	279	393	67	223
1966	6 719	5 179	2 440	1 692	675	372	158	272	322	423	68	276
1967	7 254	5 552	2 667	1 719	732	414	185	304	342	429	76	365
1968	8 296	6 426	3 114	1 882	995	435	194	339	373	478	87	376
I	7 779	6 015	2 893	1 738	951	433	164	312	350	485	74	356
II	7 615	5 927	2 858	1 727	944	398	186	307	342	435	75	320
III	8 224	6 324	3 039	1 835	1 023	427	193	346	379	483	99	376
IV	9 566	7 439	3 666	2 229	1 063	481	234	389	420	508	102	453
1969	9 463	7 370	3 765	2 133	991	481	244	393	396	512	99	424
I	8 536	6 625	3 503	1 915	764	443	201	353	378	479	80	402
II	9 500	7 463	3 799	2 093	1 087	484	247	375	390	495	102	404
III	9 516	7 359	3 737	2 122	1 016	484	244	412	414	531	111	423
IV	10 300	8 033	4 020	2 403	1 098	512	284	433	402	544	101	465

1 Excluding Eastern Bloc countries.
Source : Statistisches Bundesamt.

Table H **Merchandise Trade by Commodity Groups**
DM million, monthly averages

	1966	1967	1968	1969	1968 I	1968 II	1968 III	1968 IV	1969 I	1969 II	1969 III	1969 IV
					Imports cif							
Products of agriculture, forestry and fishing	1 182	1 095	1 107	1 232	1 054	1 104	1 066	1 203	1 201	1 395	1 151	1 182
Basic materials and semi-finished goods	2 203	2 254	2 800	3 314	2 660	2 672	2 856	3 013	3 027	3 231	3 447	3 551
Manufactured foods and tobacco	550	527	584	664	542	533	572	690	626	634	660	736
Other consumer manufactures	808	688	863	1 108	799	751	908	994	1 068	1 058	1 135	1 172
Investment goods	1 024	1 004	1 174	1 544	1 088	1 164	1 144	1 300	1 411	1 611	1 479	1 682
Other imports	289	281	237	302	190	249	293	216	281	272	315	340
Total	6 056	5 849	6 765	8 164	6 331	6 472	6 840	7 417	7 614	8 201	8 187	8 663
					Exports fob							
Products of agriculture, forestry and fishing	65	73	82	104	84	71	74	101	87	93	103	136
Basic materials and semi-finished goods	2 076	2 193	2 571	2 788	2 492	2 373	2 597	2 820	2 637	2 748	2 858	2 909
Manufactured foods and tobacco	129	164	198	233	177	188	209	219	203	239	246	243
Other consumer manufactures	682	736	855	1 063	769	753	888	1 008	965	1 035	1 108	1 143
Investment goods	3 655	3 870	4 461	5 135	4 134	4 112	4 323	5 275	4 512	5 243	5 069	5 717
Other exports	112	118	128	140	122	115	134	143	132	141	134	153
Total	6 719	7 254	8 296	9 463	7 779	7 615	8 224	9 566	8 536	9 499	9 518	10 301

Source : Statistisches Bundesamt.

Table I **The Balance of Payments**
DM million

	1966	1967	1968	1969*p*
Trade balance	11 825	21 009	22 703	19 977
Exports fob	80 557	86 950	99 445	113 473
Imports fob	68 932	66 303	76 807	93 590
Merchandise transactions abroad	200	362	65	94
Services, net	–4 994	–4 709	–3 967	–4 763
Foreign travel, net	–3 066	–2 727	–2 697	–3 658
Transportation, net	–515	–517	–646	–594
Investment income, net	–1 482	–1 783	–837	–538
Receipts from foreign troops	4 898	5 241	5 347	5 963
Other services, net	–4 829	–4 923	–5 134	–5 936
Transfers, net	–6 324	–6 434	–7 315	–8 160
Official, net	–2 871	–3 315	–4 141	–4 362
Private, net	–3 453	–3 199	–3 174	–3 798
Of which: foreign worker remittances	–2 529	–2 162	–2 153	–2 742
Current balance	507	9 866	11 421	7 054
Long-term capital, net[1]	270	–3 203	–11 513	–22 975
Official, net[1]	–1 499	–1 515	–1 366	–1 839
Private, net	–1 769	–1 690	–10 157	–21 136
Foreign investment in Germany, net	4 384	1 673	1 681	633
Portfolio investment, net	–336	–619	13	–1 208
Direct investment	3 441	2 811	1 483	1 232
Loans and credits, net	1 400	–391	320	708
Other, net	–121	–128	–135	–99
German investment abroad, net	–2 615	–3 363	–11 838	–21 769
Portfolio investment, net	–819	–1 395	–5 627	–9 311
Direct investment	–1 228	–1 042	–1 594	–2 143
Loans and credits, net	–368	–755	–4 393	–9 952
Other, net	–200	–171	–224	–363
Basic balance	777	6 663	–102	–15 921
Non-monetary short-term capital	1 952	–1 542	2 267	–364
Errors and omissions	564	15	2 935	1 671
Balance on non-monetary transactions	3 293	5 136	5 100	–14 614
Private monetary instit., short-term capital	–574	–4 823	2 455	4 325
Assets	–165	–6 010	–3 517	–2 648
Liabilities	–409	1 187	5 972	6 973
Balance on official settlements	2 719	313	7 555	10 289
Special transactions	–472	—	–500	–777
Miscellaneous official accounts	392	–8	(144)	
Change in reserves (+ = increase)[2]	2 398	494	7 180	–10 279
(*a*) Gold	–472	–257	1 246	–1 662
(*b*) Currency assets	2 147	1 572	4 080	–4 162
(*c*) Net position in IMF	723	–821	1 854	–4 455

1 Excluding special transactions.
2 Including Roosa Bonds.
p Preliminary.
Source : Deutsche Bundesbank.

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